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 SURVEY

Adversarial Evasion in Machine-Learning-Based Network Intrusion Detection: A Systematic Review, Threat Modeling, and Research Roadmap

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ABSTRACT Machine learning-based Network Intrusion Detection Systems (ML-based NIDS) have become central to modern cyber defense because they can learn complex traffic patterns and detect evolving malicious behavior beyond predefined signatures. However, their reliance on statistical feature representations also exposes them to adversarial evasion, where malicious traffic is modified at inference time to induce misclassification while preserving its operational intent. In network environments, such attacks are shaped by protocol semantics, feature dependencies, traffic timing, and functionality-preservation constraints, making adversarial evasion in ML-based NIDS distinct from adversarial examples in more continuous domains. This review synthesizes 186 studies published between 2018 and 2026—a fast-growing but fragmented body of work—using a Kitchenham-informed systematic literature review methodology. It introduces a lifecycle-aware, NIDS-specific threat model and organizes the literature into three domains—adversarial generation, robustness evaluation, and defense—developing a hierarchical taxonomy within each and conducting a two-tier critical analysis that separates domain-specific limitations from cross-cutting methodological, empirical, and operational constraints. The synthesis shows that much reported evasion success reflects diagnostic feature-space sensitivity rather than executable, deployable adversarial risk, and that many defenses demonstrate component-level robustness rather than system-level resilience. The review also develops a perturbation-realism taxonomy, ranging from feature-level manipulation to executable packet-level attacks, that clarifies when reported success corresponds to deployable risk. Finally, it derives evidence-grounded lessons learned and a future research roadmap for improving realistic evaluation, reproducible reporting, and deployable robustness in adversarial ML-based NIDS.

INDEX TERMS Network intrusion detection, adversarial examples, adversarial robustness, adversarial attacks, evasion attacks, cyber security, adversarial machine learning.

I. INTRODUCTION

Network Intrusion Detection Systems (NIDS) play a central role in protecting digital infrastructures by monitoring network traffic and identifying malicious behavior, intrusion attempts, and policy violations [1], [2]. As cyber threats become more adaptive and modern networks grow more dynamic, traditional signature-based detection is increasingly limited by its dependence on predefined attack patterns [2]. Machine learning (ML) has therefore been widely adopted in NIDS to

support data-driven and adaptive detection, allowing models to learn statistical representations of benign and malicious traffic and to recognize previously unseen attack patterns [1], [2]. Yet this same reliance on learned decision boundaries introduces a critical weakness: carefully crafted perturbations to network traffic can shift malicious flows into regions classified as benign [3].

Adversarial evasion attacks exploit this weakness at inference time by modifying malicious traffic so that it bypasses detection while preserving its harmful functionality [4]. Unlike poisoning attacks, which compromise the learning process during training, evasion attacks target deployed detectors

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and are therefore directly relevant to operational NIDS [3]. However, adversarial evasion in network intrusion detection is not a direct transfer of adversarial examples from domains such as computer vision or natural language processing. Network traffic is governed by protocol semantics, flow dependencies, timing behavior, and payload or functionality constraints; consequently, perturbations must remain feasible as traffic behavior rather than merely valid as numerical feature changes [5], [6]. This makes adversarial evasion in ML-based NIDS a distinct research problem at the intersection of cybersecurity, network measurement, and adversarial machine learning [7].

Research on adversarial attacks and defenses for ML-based NIDS has expanded rapidly, covering attack-generation techniques, robustness evaluations, and defensive mechanisms [5]. However, the resulting literature remains difficult to compare because studies differ substantially in threat assumptions, datasets, feature representations, perturbation constraints, evaluation metrics, transferability settings, and operational validation [6]. In many cases, it remains unclear whether reported attack success reflects deployable traffic-level evasion or only sensitivity to feature-space perturbations, and whether defense effectiveness generalizes beyond narrow datasets or non-standardized adversarial settings. These inconsistencies make it difficult to determine which findings indicate diagnostic model sensitivity, which demonstrate realistic adversarial risk, and which support deployable robustness. As this literature grows quickly without shared assumptions or reporting standards, consolidating it into deployment-relevant conclusions is increasingly urgent.

Existing surveys provide valuable foundations for adversarial ML in cybersecurity and intrusion detection, but several remain broad across security domains, centered on general IDS attacks and defenses, focused mainly on deep learning, or primarily descriptive in their treatment of realism and evaluation practice. The aim of this review is therefore to organize adversarial evasion research in ML-based NIDS, separate attack generation from evaluation and defense, and assess which reported results are operationally meaningful. Motivated by these considerations, this review seeks to address the following central research question:

What adversarial evasion attacks are employed for ML-based NIDS, what defense mechanisms have been proposed, and what limitations and future research opportunities emerge from current practices?

Contributions: This review makes the following key contributions to the study of adversarial evasion attacks against ML-based NIDS; the overall structure of the review is summarized in Fig. 1.

- 1) **Focused, Comprehensive, and Methodologically Transparent Evidence Base:** We synthesize 186 studies published between 2018 and 2026 within a NIDS-exclusive and evasion-centered scope, using a Kitchenham-informed systematic review protocol with explicit research questions,

multi-database search, screening criteria, and structured data extraction.

- 2) **Lifecycle-Aware ML-based NIDS Threat Model:** We develop a lifecycle-aware threat model that links adversary position, lifecycle stage, minimum capability, root vulnerability, risk profile, impact, and mitigation focus, clarifying where evasion sits within the broader ML-based NIDS attack surface.
- 3) **Three-Domain Categorization with Hierarchical Taxonomy:** We organize the literature into generation, evaluation, and defense domains and construct a hierarchical taxonomy within each domain to separate how evasive examples are produced, how robustness is measured, and how evasion impact is mitigated.
- 4) **Two-Tier Limitation Analysis and Perturbation-Realism Taxonomy:** We distinguish category-specific methodological weaknesses from cross-category systemic gaps and analyze perturbation constraints ranging from feature-space manipulation to protocol-aware, functionality-preserving, and executable packet-level attacks.
- 5) **Evidence-Grounded Lessons Learned and Future Research Roadmap:** We derive dimension-aligned lessons learned and a future research roadmap from the diagnostic synthesis, linking observed limitations to concrete directions for improving robustness evaluation, defense generalization, and operational applicability.

II. BACKGROUND

This section provides the technical foundation needed to interpret adversarial evasion in ML-based NIDS. It first summarizes how ML-based NIDS operate, then introduces adversarial examples and transferability, and finally outlines the main generic attack families adapted to NIDS together with the domain-specific constraints that make network evasion different from adversarial attacks in continuous domains such as computer vision.

A. OVERVIEW OF ML-BASED NIDS

Intrusion detection refers to monitoring and analyzing network or system activity to identify suspicious behavior associated with potential security threats [1], [2]. IDS solutions are commonly categorized as host-based or network-based. This review focuses on Network Intrusion Detection Systems (NIDS), which inspect traffic across network layers to detect malicious behavior such as scanning, malware communication, denial-of-service, or other attack traffic [1]. Traditional NIDS generally rely on either signature-based detection, which matches traffic against known attack patterns, or anomaly-based detection, which identifies deviations from normal behavior [2], [8]. Signature-based methods are effective for known threats but limited against zero-day attacks, while anomaly-based methods can detect novel behavior but often suffer from false positives [1], [8].



FIGURE 1. Overview of the review structure and the main topics covered in each section.

ML-based NIDS extend anomaly-based detection by learning statistical relationships from network traffic rather than relying on hand-written rules. Raw traffic is first captured from packet captures, flow exporters, or logs and transformed through preprocessing and feature extraction into a structured representation; a model is then trained on this representation, evaluated, and deployed to classify live traffic and raise alerts [2]. The learning paradigm depends on how much labeled traffic is available: supervised models such as Support Vector Machines, Decision Trees, Random Forests, and Multi-Layer Perceptrons are effective when labeled attacks are plentiful, whereas semi-supervised and unsupervised approaches address the more common case in which malicious examples are scarce. Deep architectures such as Convolutional, Recurrent, and Long Short-Term Memory networks reduce the need for manual feature engineering by learning spatial or temporal structure directly from traffic [1], [2], [8]. This learned flexibility is also the source of the vulnerability examined in this review: a model that infers its own decision boundary can be pushed across that boundary by carefully chosen input perturbations.

B. ADVERSARIAL EXAMPLES IN ML-BASED NIDS

ML-based NIDS are vulnerable to adversarial evasion because malicious traffic can be modified to cross a learned decision boundary while preserving its attack intent [3], [5]. In this review, an adversarial example refers to a network traffic instance whose feature, flow, timing, or

packet-level representation has been modified so that the detector misclassifies it as benign while the underlying malicious functionality remains intact [5].

Formally, adversarial perturbations are often modeled as constrained optimization problems [4]. Given a classifier f_θ , an input x with label y , a loss function L , and a perturbation budget ϵ , an adversarial instance can be expressed as:

$$x^{adv} = x + \delta, \quad \text{s.t. } \|\delta\|_p \leq \epsilon, \quad f_\theta(x^{adv}) = y_{\text{benign}}. \quad (1)$$

In ML-based NIDS, x usually represents a vectorized packet, flow, or traffic record, while δ represents strategic changes to traffic-related features. Unlike image-domain perturbations, however, NIDS perturbations must respect network-specific constraints such as valid feature ranges, protocol semantics, temporal consistency, and malicious-functionality preservation.

Transferability is another important property of adversarial examples. It refers to the ability of perturbations crafted against one model to remain effective against another model, even when the target architecture or training process differs [3]. In NIDS, transferability enables black-box or gray-box evasion because an attacker may train a substitute detector, craft evasive samples against it, and reuse them against the target system without direct access to its internal parameters. The practical meaning of such transfer depends on how closely the surrogate and target share feature extraction, preprocessing, and traffic representation assumptions.

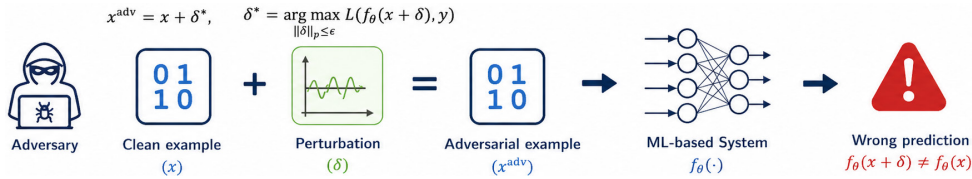


FIGURE 2. Adversarial example generation pipeline.

C. ADVERSARIAL METHODS AND NIDS-SPECIFIC CONSTRAINTS

Most adversarial methods adapted to NIDS originate from broader adversarial ML research rather than from network-security-specific design. These methods provide useful baselines for testing detector sensitivity, but their direct application to NIDS is limited when they treat packet or flow records as generic feature vectors. Table 1 summarizes the main attack families, their representative methods, and their typical NIDS-specific limitations.

The key challenge in NIDS is that adversarial perturbations must satisfy constraints that are weaker or absent in many continuous-domain settings. Feature-space changes may not map back to valid packets or replayable traffic; protocol fields and flow statistics are interdependent; and traffic modifications must preserve malicious functionality while remaining feasible under real deployment conditions [5], [6], [7]. These constraints are introduced here only as background; the perturbation-constraints and realism analysis in Section IX examines them in detail as a central limitation of current adversarial NIDS research.

Operational factors further affect how adversarial methods should be interpreted in network security. Dataset staleness, distribution drift, encrypted traffic, inline versus passive monitoring, processing overhead, and uncertain transferability can all change whether an attack that succeeds in a benchmark setting remains meaningful in deployment [18]. Accordingly, generic adversarial attacks should be viewed as useful diagnostic tools, while stronger operational claims require evidence that perturbations are valid, functional, and feasible in realistic network environments, as analyzed in Section IX.

III. COMPARISON WITH OTHER SURVEYS

Prior surveys on adversarial machine learning in cybersecurity have established important foundations for understanding attacks, defenses, and threat models across domains such as malware detection, phishing, IoT security, anomaly detection, and intrusion detection. However, because these surveys differ in scope and methodological orientation, they also leave different gaps for NIDS-focused adversarial evasion research. Some provide broad cross-domain taxonomies, but necessarily treat NIDS as one application among many. Others focus more directly on IDS or NIDS, but vary in their treatment of evasion realism, methodological transparency, structured synthesis, and evidence-grounded future directions. This section positions the present review relative to these efforts and clarifies the specific gap it addresses.

A. POSITIONING OF PRIOR SURVEYS

The existing literature can be understood along a spectrum from broad adversarial-ML surveys to NIDS-centered reviews. Broad surveys, including Rosenberg et al. [19], Martins et al. [4], Khazane et al. [20], and Ibitoye et al. [7], are valuable because they situate adversarial learning within a wider cybersecurity landscape. They compare attacker goals, knowledge assumptions, attack phases, and defensive strategies across several domains. This breadth is useful for conceptual framing, but it also limits the depth with which these works can analyze the specific constraints of adversarial evasion in ML-based NIDS, where feature-space manipulation, protocol validity, functionality preservation, and deployment feasibility are central concerns.

More focused IDS and NIDS surveys move closer to the present review's scope. He et al. [5] provide a detailed DL-oriented analysis of vulnerabilities across the NIDS pipeline, while Alotaibi and Rassam [21] organize adversarial attacks and defenses using threat-model and defense-oriented categories. Saini et al. [22] and Ennaji et al. [18] further expand the discussion of attacks, countermeasures, semantic validity, and structured-data constraints. These reviews make important contributions, yet the field still lacks a synthesis that combines NIDS-exclusive focus, evasion-centered analysis, lifecycle-aware threat modeling, systematic extraction, perturbation-realism analysis, and a roadmap derived directly from the reviewed evidence.

Taken together, prior surveys reveal three recurring gaps. First, evasion is often discussed alongside poisoning, model extraction, and other adversarial settings, which can obscure the distinct assumptions and deployment conditions of evasion attacks. Second, the realism of adversarial examples is not always analyzed in terms of whether feature-space perturbations can be translated into protocol-valid and functionality-preserving network behavior. Third, many surveys remain primarily descriptive, with limited integration between attack generation, robustness evaluation, defense design, methodological weaknesses, lessons learned, and future research priorities. Table 2 summarizes how representative surveys address these dimensions. The symbols indicate explicit coverage of each comparison dimension; they are not intended as an overall quality ranking.

B. HOW THIS REVIEW ADDRESSES THE REMAINING GAPS

The present review addresses these gaps through a focused but integrated design. Its first contribution is a **NIDS-exclusive and evasion-centered scope**. Rather than treating

TABLE 1. Generic adversarial attack families adapted in ML-based NIDS studies.

Attack family	Representative methods	Core idea	Typical access	NIDS-specific limitation
Gradient-based attacks	FGSM, BIM/I-FGSM, PGD [9]–[11]	Follow the loss gradient to move the input toward misclassification under a perturbation budget	White-box	Often operate in feature space and require gradient access, which may be unrealistic for deployed NIDS
Optimization-based attacks	L-BFGS, DeepFool, C&W [12]–[14]	Search for low-distortion perturbations that force a target or untargeted misclassification	White-box	May minimize numerical distortion without ensuring protocol validity or traffic executability
Saliency-based attacks	JSMA [15]	Select highly influential features using model sensitivity or saliency information	White-box or gray-box	Sparse feature edits may not correspond to manipulable traffic fields or valid packet behavior
Query-based black-box attacks	ZOO, HopSkipJump [16], [17]	Estimate gradients or decision boundaries using model queries rather than internal gradients	Black-box	Query cost, feedback availability, and probing detectability affect practicality in deployed NIDS
Transfer-based attacks	Surrogate-model attacks [3]	Craft adversarial examples on a substitute model and transfer them to the target detector	Gray-box or black-box	Success depends on feature-pipeline alignment, surrogate fidelity, and shared preprocessing assumptions

TABLE 2. Comparison of surveys on adversarial ML in NIDS (✓ = Explicit, Δ = Partial, ✗ = Not Addressed).

Survey	Year	NIDS Focus	Evasion Emphasis	Threat Model Specificity	SLR Protocol	Literature Structuring	Critical Analysis	Lessons Learned	Future Directions
Martins et al. [4]	2020	Δ	✗	✗	Δ	Δ	✗	✗	✗
Rosenberg et al. [19]	2022	Δ	Δ	✓	✗	✓	✓	Δ	✓
He et al. [5]	2023	✓	✓	✓	✗	✓	✓	✓	✓
Alotaibi & Rassam [21]	2023	✓	Δ	Δ	✗	✓	Δ	Δ	Δ
Khazane et al. [20]	2024	Δ	Δ	Δ	✗	✓	Δ	Δ	✗
Saini et al. [22]	2024	✓	Δ	Δ	Δ	✓	✓	Δ	Δ
Ibitoye et al. [7]	2025	Δ	Δ	✓	✗	✓	✓	✓	✓
Ennaji et al. [18]	2025	✓	✓	Δ	✗	✓	✓	Δ	Δ
This Review	2026	✓	✓	✓	✓	✓	✓	✓	✓

Note. The symbols indicate the level of explicit coverage of each comparison dimension, not an overall ranking of survey quality. A checkmark (✓) indicates that the dimension is explicitly and substantially addressed; a triangle (Δ) indicates partial or indirect coverage; and a cross (✗) indicates that the dimension is absent or not clearly reported. For the *SLR Protocol* column, ✓ means the survey reports an explicit search strategy, eligibility criteria, corpus definition, data-extraction schema, and quality-control procedure. ✗ means no such protocol is reported—no search string, database list, eligibility criteria, or study-selection flow—and is not a judgment of survey quality.

evasion, poisoning, extraction, and other adversarial settings as interchangeable, the review concentrates on deployment-stage evasion as the adversarial mechanism most directly connected to malicious traffic manipulation against operational NIDS. At the same time, the broader lifecycle threat model is retained to explain how upstream and downstream weaknesses can shape evasion feasibility.

The second contribution is a **lifecycle-aware ML-based NIDS threat model**. By connecting adversary position, lifecycle stage, minimum capability, root vulnerability, risk profile, impact, and mitigation focus, the threat model clarifies where evasion sits within the broader ML-based NIDS attack surface. This framing helps distinguish diagnostic white-box assumptions from gray-box and black-box settings that are more plausible under deployed external-adversary conditions.

The third contribution is a **three-domain taxonomy of generation, evaluation, and defense**. This organization separates studies that propose adversarial attack-generation techniques from studies that evaluate detector robustness or design defensive mechanisms. The separation is important because each domain answers a different question: how evasive examples are produced, how robustness is measured, and how evasion impact is reduced. Treating these domains

separately improves interpretability and avoids collapsing distinct assumptions into a single adversarial-ML catalogue.

The fourth contribution is a **realism-oriented critical synthesis**. The review connects perturbation constraints, feature-to-traffic validity, methodological reproducibility, operational feasibility, and lifecycle limitations to the evidence reported across the literature. These analyses support the lessons learned and future research roadmap, ensuring that the proposed directions follow from observed methodological and operational gaps rather than from generic calls for more robust models.

IV. METHODOLOGY

This review adopts a structured systematic literature-review methodology informed by Kitchenham’s planning, conducting, and reporting stages [23]. The aim was to provide a transparent and reproducible synthesis of adversarial evasion research in ML-based NIDS without treating heterogeneous studies as directly comparable statistical observations. The protocol defined the research questions, search sources, search query, eligibility criteria, data-extraction fields, and synthesis dimensions used to organize the reviewed literature.

TABLE 3. Research questions and corresponding sections addressed in the review.

RQ	Research question	Addressed in
RQ1	What are the primary limitations of current adversarial evasion attack generation methods for ML-based NIDS?	Sections VI and IX
RQ2	What characteristics are required for practical and effective adversarial example generation in NIDS contexts?	Sections VI and IX
RQ3	How is NIDS robustness typically evaluated under adversarial settings, and what methodological gaps remain?	Sections VII and IX
RQ4	What mitigation and defense strategies have been proposed, and what are their comparative advantages and weaknesses?	Section VIII
RQ5	What design and evaluation principles are essential for developing resilient ML-based NIDS against adversarial evasion?	Section IX

A. REVIEW DESIGN AND RESEARCH QUESTIONS

The review synthesizes studies on adversarial evasion in ML- and DL-based NIDS across three analytical dimensions: adversarial attack generation, robustness evaluation, and defense. These dimensions structure the taxonomy and guide the later discussion of methodological and operational gaps. The five research questions operationalize the central research question stated in the Introduction and map these dimensions to the main analytical sections of the review. Table 3 summarizes the research questions and the sections in which they are addressed.

B. SEARCH STRATEGY AND STUDY SELECTION

A literature search was conducted across major bibliographic databases, including Scopus, Web of Science, IEEE Xplore, ACM Digital Library, and ScienceDirect. To capture recent and preprint studies, supplementary searches were performed using Google Scholar and arXiv. The search targeted studies published or made available between May 2018 and June 2026. The search combined terms related to adversarial machine learning and the NIDS application domain using the core query (*“adversarial machine learning” OR “adversarial attack”*) AND *“network intrusion detection”*, with additional variants applied during screening to capture related terms such as *adversarial example*, *evasion attack*, *robustness*, *adversarial defense*, *deep learning*, and *network traffic*.

Records retrieved from database and supplementary searches were consolidated, checked for duplicates and clearly irrelevant results, and then screened by title, abstract, and full text against the eligibility criteria. After screening and eligibility-based quality control, 186 studies were retained for detailed extraction and synthesis. A complete summary of the 186 studies included in the reviewed corpus is provided in Appendix Table 11. These 186 studies form the reviewed corpus analyzed in Sections VI–VIII; further references cited throughout the paper support the background, methodology, and discussion but are not part of the reviewed corpus. The final corpus comprised 63 generation studies, 48 evaluation studies, and 75 defense studies. Figure 3 summarizes the

Kitchenham-informed study-selection and synthesis workflow used to construct the reviewed corpus.

C. INCLUSION, EXCLUSION, AND QUALITY CONTROL

Studies were included if they met three conditions: they addressed an ML- or DL-based NIDS context using network-traffic data; they examined at least one adversarial dimension relevant to this review (attack generation, robustness evaluation, or defense); and they reported extractable empirical information such as the dataset, model, attack or defense mechanism, evaluation setting, and measurable outcomes. Studies were excluded if they focused on non-NIDS domains, lacked an ML/DL component, did not involve an adversarial context, or reported no empirical evaluation.

Rather than applying a separate numerical quality-appraisal score, quality control was embedded in these eligibility criteria: each retained study had to provide sufficient methodological detail to support structured extraction and comparison, and was screened for relevance, empirical adequacy, and reporting completeness. This approach suits the present review because the retained studies vary substantially in datasets, models, attack assumptions, perturbation constraints, evaluation metrics, and defense objectives, making a single numerical quality score difficult to apply consistently across the corpus.

Because the review was conducted by a single author, a second-pass self-audit was performed to reduce screening bias. A random subsample of 30 excluded records was re-screened against the inclusion and exclusion criteria, and all sampled records were confirmed as correct exclusions. Even so, single-reviewer screening is inherently more prone to selection bias than independent dual screening; the self-audit mitigates but does not eliminate this residual risk.

D. DATA EXTRACTION AND THEMATIC SYNTHESIS

A structured extraction and coding schema ensured consistency across the retained studies. Extraction focused on conceptual and methodological attributes rather than pooling raw numerical performance values: for each paper, bibliographic details, research objective, dataset, detection model, attack or defense mechanism, evaluation setting, reported findings, and limitations were recorded, and each study was coded into one of the three analytical dimensions used in this review (*adversarial generation*, *evaluation*, or *defense*). Table 4 summarizes the schema.

Quality control was implemented qualitatively rather than through a separate numerical scoring scheme. During screening and extraction, each retained study was checked for NIDS relevance, adversarial relevance, dataset adequacy, methodological clarity, empirical adequacy, and reproducibility signal. Specifically, studies had to involve ML/DL-based network intrusion detection or network-traffic classification, address adversarial evasion generation, robustness evaluation, or defense, use network-traffic data or a clearly relevant NIDS benchmark, and report sufficient methodological and

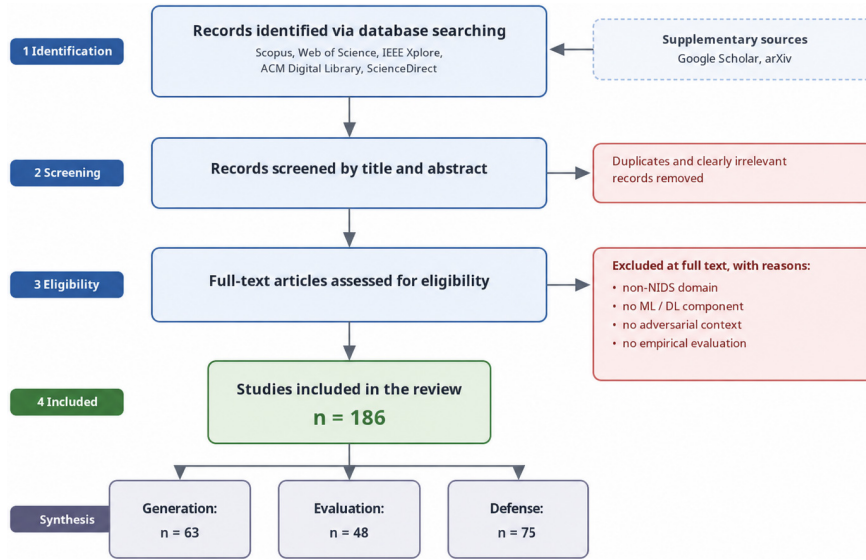


FIGURE 3. Kitchenham-informed study-selection and synthesis workflow used to construct the reviewed corpus. Quality control was embedded in the eligibility criteria.

TABLE 4. Data extraction and coding schema.

Attribute	Description
Bibliographic details	Authors, year, venue, and publication source.
Research objective	Main goal, contribution, or problem addressed.
Research dimension	Classification as generation, evaluation, or defense.
Dataset	Network dataset, benchmark, or traffic source used.
Detection model	ML/DL model or architecture evaluated.
Attack mechanism	Evasion method, perturbation strategy, or adversarial setting.
Adversary knowledge	White-box, gray-box, black-box, transfer-based, or query-based setting.
Perturbation constraint	Feature-space, protocol-aware, functionality-preserving, packet-level, or realism-related constraint.
Defense mechanism	Robustness, detection, filtering, purification, ensemble, or hybrid defense strategy.
Evaluation metrics	Accuracy, detection rate, F1-score, attack success rate, robustness, or computational cost.
Reported findings	Main empirical outcomes and qualitative insights.
Reported limitations	Constraints or gaps reported by the authors or inferred from the study design.

empirical detail to support structured extraction. These checks were used to guide inclusion, extraction, and interpretation rather than to rank studies numerically, because the reviewed corpus varied substantially in datasets, models, adversary assumptions, evaluation settings, and reported metrics.

Because the reviewed studies differ substantially in datasets, feature representations, adversary assumptions, perturbation constraints, model architectures, and metrics, statistical meta-analysis was not appropriate. The review instead used structured thematic synthesis: studies were grouped by primary contribution and analyzed within subcategories reflecting their mechanisms, assumptions, and limitations. This enabled comparison across generation, evaluation, and defense studies while preserving the methodological differences that shape how their results should be interpreted.

V. THREAT MODELING FOR ML-BASED NIDS

Machine learning expands the attack surface of NIDS because security failures can arise across the data, model, deployment, and update pipeline rather than only at detection time [3], [7]. Existing adversarial-ML surveys and IDS-oriented taxonomies commonly classify attacks by adversary knowledge, objective, attack strategy, or mitigation type [18], [21]. Unlike prior surveys, the threat model in this review is used as a lifecycle mapping framework that links each threat class to the ML-based NIDS lifecycle stage, adversary position, minimum capability, root vulnerability, risk profile, impact, and mitigation focus.

The scope of the threat model is broader than adversarial evasion because evasion is shaped by upstream and downstream lifecycle assumptions. Poisoned data, compromised updates, leaked model artifacts, excessive feedback, insecure dependencies, or fragile parsers can all affect whether evasion is feasible, persistent, or detectable. Non-evasion threats are included only to contextualize the lifecycle threat model and are not treated as part of the reviewed evasion corpus; the corpus synthesis remains centered on evasion-relevant generation, evaluation, and defense studies. The full matrix therefore provides the security context for ML-based NIDS, while the remainder of this review focuses on adversarial evasion because it is the deployment-stage threat most directly aligned with external adversarial manipulation of malicious traffic.

A. THREAT CLASSES AND MATRIX SCOPE

The threat model distinguishes persistent lifecycle threats, such as poisoning, backdooring, tampering, model extraction, and supply-chain compromise, from transient deployment-stage threats, such as evasion and fuzzing. Non-evasion threats are included because they affect the integrity, confidentiality, availability, and maintainability of

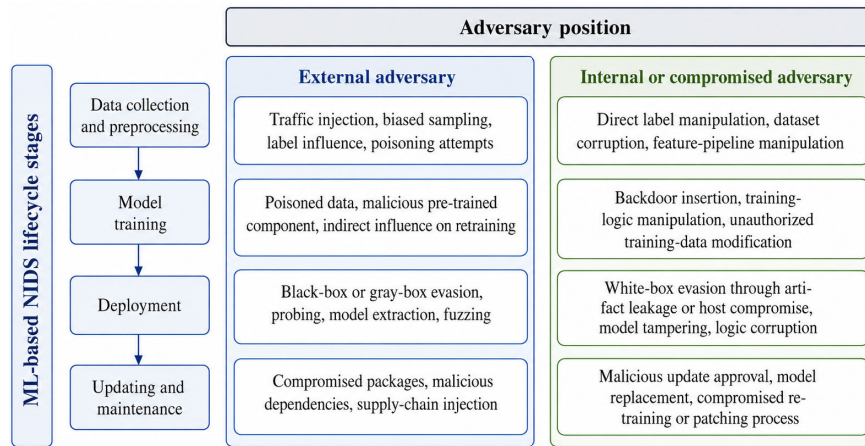


FIGURE 4. Lifecycle-adversary view of the ML-based NIDS threat surface. The figure positions threat classes by lifecycle stage and adversary position, showing how upstream data and training threats, deployment-stage evasion, and update or supply-chain compromise shape the security context of ML-based NIDS.

ML-based NIDS. Data poisoning corrupts the learning pipeline by injecting, relabeling, or manipulating training examples [22]. Model backdooring implants hidden trigger behavior through poisoned data, malicious pre-trained components, or compromised updates [6]. Model tampering modifies deployed artifacts or preprocessing logic after training [7]. Third-party compromise exploits dependencies, packages, libraries, feature extractors, or update mechanisms [19]. Fuzzing uses malformed or edge-case inputs to expose parser, feature-extraction, or runtime failures [19]. Model extraction reconstructs a surrogate detector from repeated queries and observable outputs, enabling later transfer-based or gray-box evasion [6].

Evasion threats are separated into white-box, gray-box, and black-box settings because the attacker's visibility strongly determines both realism and attack strength. White-box evasion assumes access to architecture, weights, or preprocessing logic, making it useful as a diagnostic upper-bound setting, but less realistic as an external deployed threat unless artifacts are leaked or the host is compromised [21], [24]. Gray-box evasion relies on partial knowledge, such as feature-extraction logic or input format, and supports transfer-based attacks through substitute models [21]. Black-box evasion treats the detector as opaque and depends on probing, observable alerts, confidence scores, or indirect feedback [21]. Table 5 consolidates these threats into a compact lifecycle matrix.

B. ATTACK CHARACTERIZATION DIMENSIONS

Each threat in the matrix is characterized through feasibility, persistence, impact, detectability, and minimum capability. Mitigation logic is treated separately through the vulnerability-root-cause analysis in Table 6. These dimensions provide the explicit criteria behind the ratings in Table 5.

Minimum capability refers to the least adversary access, knowledge, or control required to execute the threat, such as traffic injection, query access, partial knowledge of the feature

pipeline, model-artifact access, or compromise of an update channel.

Feasibility refers to how realistic it is to launch an attack under given assumptions. We distinguish three levels: *theoretical*, relying on strong or currently unrealistic assumptions about attacker capability; *conditional*, viable only under specific environmental conditions or vulnerabilities; and *practical*, implementable in plausible deployment scenarios with realistic adversary resources.

Persistence describes whether the attack effect is short-lived, such as transient evasion, or long-lasting, such as poisoning that degrades all future detections [22].

Impact captures the consequence of a successful attack, categorized into four outcomes [21]: *loss of model integrity* (corruption of decision boundaries, producing degraded performance and stealthy, selective evasion); *loss of system integrity* (unauthorized modification or subversion of the NIDS, enabling total compromise, attacker control, or denial-of-service); *loss of model confidentiality* (disclosure of model parameters, architecture, or training data); and *loss of system availability* (disruption of the NIDS operational state through resource exhaustion or induced crashes).

Detectability reflects the likelihood that monitoring or security controls observe the attack, ranging from *highly stealthy* threats such as clean-label poisoning or subtle perturbations to *easily observable* ones such as fuzzing or brute-force probing [22].

Together, these dimensions provide an integrated view of how attack vectors emerge and interact across the ML-based NIDS pipeline. The threat matrix in Table 5 applies these dimensions per threat class, while Table 6 maps the underlying vulnerabilities to mitigation logic.

C. MATRIX-LEVEL INSIGHTS

The matrix shows that operational risk diverges from the attack settings most commonly emphasized in adversarial-ML experimentation. White-box evasion is valuable for stress

TABLE 5. Integrated threat matrix for ML-based NIDS. Risk profile is reported as feasibility / persistence / detectability. Mitigation logic is given per root cause in Table 6.

Threat class	Stage	Adversary	Min. capability	Root vulnerability	Risk profile	Impact
Data poisoning	Data collection, training	External, internal	Inject/alter traffic, labels, or training data	Untrusted data pipeline	Practical / long-term / stealthy	Model integrity
Model backdooring	Training, updating	External, internal	Influence training data or update artifacts	Untrusted pipeline; insecure supply chain	Conditional / long-term / stealthy	Model integrity
Model tampering	Deployment	Internal	Modify model or preprocessing logic	Weak artifact protection	Conditional / long-term / observable	System integrity
White-box evasion	Deployment	Internal, leaked-access	Access weights, gradients, or pipeline	Weak artifact protection	Theoretical (deployed) / transient / stealthy	Model integrity
Gray-box evasion	Deployment	External	Infer feature extractor or input format	Excessive transparency & feedback	Practical / transient / stealthy	Model integrity
Black-box evasion	Deployment	External	Probe via alerts, labels, or scores	Excessive transparency & feedback	Practical / transient / observable	Model integrity
Model extraction	Deployment	External, internal	Query repeatedly, observe outputs	Excessive transparency & feedback	Practical / long-term / observable	Model confidentiality
Third-party compromise	Updating, deployment	External	Compromise libraries, models, or dependencies	Insecure supply chain & updates	Practical / long-term / stealthy	System integrity
Fuzzing	Deployment	External	Send malformed or edge-case traffic	Insufficient robustness & monitoring	Practical / transient / observable	System availability

testing because it exposes an upper bound on model sensitivity, but it remains theoretical as a deployed external threat unless the adversary obtains model artifacts or compromises the host. By contrast, gray-box and black-box evasion are more plausible deployment-stage threats because they require partial or indirect visibility rather than full model access. Persistent lifecycle threats, including poisoning, backdooring, model extraction, and third-party compromise, can convert short-term evasion risk into long-term degradation, future transferability, or system-level compromise. This mismatch implies that ML-based NIDS security should not be assessed only by the strength of white-box evasion results; it should also account for the lifecycle conditions that make attacks feasible, persistent, and difficult to detect.

D. VULNERABILITY ROOT CAUSES AND MITIGATION LOGIC

The threats in Table 5 are enabled by recurring vulnerability categories rather than isolated implementation mistakes. Table 6 summarizes these root causes and the mitigation logic they imply. Unlike the attack-indexed threat matrix, this table is vulnerability-indexed: it shows that one weakness can enable several threats, and one threat can depend on several weaknesses.

These root causes are interconnected. A poisoning attack may exploit an untrusted data pipeline while relying on weak monitoring to remain undetected; model extraction may enable later gray-box or transfer-based evasion; and supply-chain compromise can introduce backdoors that appear only after deployment. This many-to-many structure motivates defense-in-depth: ML-based NIDS protection must combine data governance, artifact protection, interface control, supply-chain security, monitoring, and lifecycle validation rather than relying on a single defensive layer.

E. IMPLICATIONS FOR THE REVIEW

This threat model frames the remainder of the review by distinguishing the full ML-based NIDS security lifecycle

from the narrower evasion focus of the surveyed literature. The full lifecycle matrix explains how threats can emerge before, during, and after deployment. The rest of the review then focuses on adversarial evasion because it is the most direct deployment-stage mechanism through which external attackers attempt to modify malicious traffic and bypass detection.

Within this scope, attack-generation studies ask how evasive traffic can be constructed under different knowledge and feasibility assumptions; evaluation studies ask how detectors fail under existing evasion methods; and defense studies ask how evasion impact can be reduced or contained. The cross-cutting discussion later returns to the lifecycle issues surfaced here, including perturbation realism, threat-model feasibility, reporting transparency, and operational validation. The threat model therefore provides the security baseline for interpreting which evasion claims are diagnostic, which are practically plausible, and which depend on broader lifecycle weaknesses.

VI. ATTACK GENERATION TECHNIQUES

This section reviews studies that develop adversarial attack generation techniques for ML-based NIDS. These works aim to craft malicious network traffic that evades detection while preserving operational intent, semantic validity, and, where possible, protocol-level feasibility. Unlike adversarial examples in continuous domains such as computer vision, evasion in network intrusion detection must account for discrete feature spaces, protocol dependencies, temporal behavior, and functional malicious activity.

The reviewed literature is organized according to the dominant mechanism used to generate adversarial traffic: surrogate-based and query-efficient black-box approximation, domain-constrained manipulation, gradient-free search, explanation-guided perturbation, distributional generation, sequential decision-making, and gradient-based optimization. This organization highlights the field's progression from abstract feature-space perturbations toward more realistic,

TABLE 6. Root vulnerabilities, the threat classes they enable, and the corresponding mitigation logic for ML-based NIDS. A single weakness enables several threats, and threats such as model backdooring recur across multiple root causes, reflecting a many-to-many structure that motivates layered defense.

Root vulnerability	Enabled threats	Mitigation logic
Untrusted data pipeline	Data poisoning; model backdooring	Data provenance validation, traffic sanitization, label verification, training-data access control, robust statistics, and outlier detection [5], [22], [25]
Weak model and artifact protection	Model tampering; white-box evasion; backdoor activation	Artifact signing, secure storage and boot, access control, hashing, and runtime integrity monitoring [7]
Insecure supply chain and updates	Third-party compromise; model backdooring; malicious updates	Dependency vetting, package and model provenance checks, pre-deployment update validation, patching, and sandboxing of untrusted modules [7], [19], [22]
Excessive transparency and feedback	Black-box evasion; gray-box evasion; model extraction	Query rate limiting, confidence-score suppression, reduced feedback, extraction-pattern monitoring, and randomized or decoy responses [3], [19]
Insufficient robustness and monitoring	Fuzzing; persistent compromise; drift exploitation	Adversarial testing, parser hardening, input validation, anomaly and drift detection, runtime recovery, and lifecycle-aware validation [6]

constraint-aware, and operationally meaningful evasion attempts.

A. BLACK-BOX & TRANSFERABLE ATTACKS

Black-box and transferable attacks are more deployment-plausible than white-box attacks, but their realism depends on query budget, feedback availability, and the attacker's knowledge of the target NIDS feature-extraction and preprocessing pipeline.

One line of work approximates the target detector through surrogate modeling and transfers adversarial perturbations to the victim system. Saliency-guided iterative FGSM and mutual-information-guided attacks select influential features after approximating the victim's feature pipeline or training substitute DNNs [26], [27]. Later methods link this surrogate logic to traffic-aware manipulation: label-only attacks use protocol-aware morphing, FAT maps feature-space changes to packet length and timing, and graph-based evasion uses a shadow GNN with Random Walk with Restart to identify influential nodes for structural perturbation [28], [29], [30]. Simulation-based workflows further connect black-box generation with adversarial retraining, showing how attack construction and hardening can be studied in the same pipeline [31].

Other approaches reduce full model extraction by exploiting limited feedback or reusable perturbations. Manifold-based query methods estimate anomaly-detector thresholds from clustered responses, while nearest-neighbor guidance moves perturbations toward benign regions [32], [33]. Feedback-limited vulnerability assessment uses silent probing, change-point detection, and causality analysis to identify sensitive traffic features with minimal interaction [34]. Transferability-oriented methods improve cross-model reuse through linearized surrogate boundaries or universal perturbations trained over multiple substitute models [35], [36].

Takeaway. Black-box and transferable attacks test evasion under restricted attacker knowledge rather than idealized full access. Their practical appeal is also their source of uncertainty: the less the attacker observes, the more success depends on surrogate fidelity, feature-pipeline alignment, and the quality of limited feedback. Thus, black-box evasion is realistic as a threat model but inherently approximate as an attack mechanism.

B. DOMAIN-CONSTRAINED & REALISTIC ATTACKS

Domain-constrained and realistic attacks address a core weakness of generic evasion: a perturbation that fools a detector in feature space may still be invalid, non-executable, or unable to preserve malicious behavior in real traffic. These methods embed network semantics, feature dependencies, protocol rules, and functionality-preserving assumptions into attack generation. Rather than treating all features as equally manipulable, they ask which traffic attributes can be changed, how they depend on one another, and whether the resulting manipulation can exist in a real network environment.

Some studies impose feasibility directly in feature space. A2PM encodes coherent multi-feature swaps and bounded integer edits [37]. Related work embeds feature-group feasibility into Carlini-Wagner optimization, partitions features into manipulable, dependent, and non-manipulable sets, or amplifies existing attacks only within validity-preserving bounds [38], [39], [40]. Customized UAP extends this logic to reinforcement-learning-based intrusion detection by integrating network-feature rules, feature dependencies, and a correlation-guided loss [41]. These studies collectively shift attack generation from unconstrained perturbation toward explicit modeling of what an adversary can plausibly modify [42], [43].

A more operational line of work moves from abstract features to traffic behavior, packet transformations, timing, payloads, host actions, or graph structures. Host-space perturbations avoid direct editing of feature vectors, PCAP traces, or NetFlows by modifying commands or actions on controlled hosts [44]. Other methods manipulate packet content, payloads, timing, segmentation, or flow characteristics while preserving semantic bounds, including ANT, Luer Mihou, transformation-based attacks, Linux traffic-control shaping, and TANTRA [45], [46], [47], [48], [49]. Structural and protocol-specific attacks extend this realism spectrum through constrained graph perturbations, GIANT's adversarial flow injection, Modbus TCP jitter and padding [50], [51], [52]. These studies show that realism is not binary: attacks may be feature-valid, packet-valid, graph-valid, protocol-valid, or host-executable.

Takeaway. Domain-constrained and realistic attacks make evasion claims more operationally meaningful by testing whether perturbations survive the constraints imposed by real

traffic generation and representation. Their main trade-off is locality: the more carefully an attack models a dataset, protocol, feature extractor, or deployment setting, the harder it becomes to generalize elsewhere. Realism is gained by narrowing the attack space, but that narrowing can also limit transferability across NIDS environments.

C. EVOLUTIONARY & HEURISTIC ALGORITHMS

Evolutionary and heuristic algorithms frame adversarial evasion as a search problem rather than a gradient-optimization problem. They are useful in ML-based NIDS when traffic features are discrete, constraints are difficult to differentiate, or the attacker has only black-box access. Candidate perturbations are generated, evaluated, and refined through a fitness objective that can combine evasion success, edit cost, validity constraints, and interpretability.

GA optimizes packet-sequence modifications through selection, crossover, and perturbation, while packet-saliency-guided search reduces unnecessary edits by prioritizing influential packet positions [53]. Other studies compare genetic algorithms and particle swarm optimization across classifiers, replay search-generated flows in live networks, constrain payload substitution to preserve token validity, or use decision-tree feedback to evolve interpretable evasive samples [54], [55], [56], [57]. The common contribution is not a specific heuristic, but the ability to adapt search objectives to discrete traffic manipulations and black-box constraints.

Hybrid search methods combine heuristic optimization with learned traffic distributions. In the GAN+PSO framework, the GAN learns benign traffic patterns while particle swarm optimization searches for evasive perturbations within that learned space [58]. This separates distribution learning from perturbation optimization, but the result remains shaped by the fitness function, search budget, and whether the learned space captures valid traffic behavior.

Takeaway. Evolutionary and heuristic attacks are strongest where gradients are unavailable or unsuitable, because search can combine attack success with edit cost, constraints, and interpretability. Their flexibility comes at a price set by the search budget, parameter choices, and fitness design. Search buys expressiveness, but the more hand-crafted the objective becomes, the less confidently the result generalizes beyond the benchmark setting.

D. EXPLAINABILITY-GUIDED ATTACKS

Explainability-guided attacks repurpose interpretation methods as tools for adversarial manipulation. Feature-attribution scores are used not only to explain detector decisions but also to identify traffic attributes that are most valuable to perturb. This gives the attacker a more directed strategy than unguided search or broad gradient updates.

Integrated Gradients has been used to identify influential features and map perturbations to feasible problem-space edits [59]. KernelSHAP extends this idea to black-box settings, while DeepSHAP-guided masked projected-gradient

descent and SHAP-guided perturbation prioritize high-impact traffic fields to produce sparser attacks [60], [61], [62]. ETA embeds explanation-based feature selection into a transfer-based pipeline with zeroth-order gradient estimation and an optimized substitute-detector ensemble [63]. Across these studies, explanations reduce the perturbation search space but make attack quality dependent on attribution stability.

Takeaway. Explainability-guided attacks expose the dual use of interpretability: attribution signals can explain detector decisions, but they can also steer evasion. Their advantage is precision, since perturbations focus on features that appear decision-critical. Their fragility lies in the explanation signal itself, which may shift across models, methods, and traffic representations.

E. GENERATIVE MODELS

Generative models approach adversarial evasion as a distribution-learning problem. Instead of directly optimizing individual feature changes, they learn patterns from benign or less detectable traffic and use them to generate evasive feature vectors, flow records, or perturbations that appear closer to normal traffic. Only a smaller subset of studies attempts packet-level or traffic-level modification, and such claims require validation of protocol validity, executability, and malicious-functionality preservation. This makes generative approaches potentially attractive in black-box settings because evasion can be driven by learned traffic similarity rather than direct model access.

GAN-based attacks form the largest group. MACGAN uses multiple surrogates for cross-family evasion, while DEMGAN improves diversity through multiple generators [64], [65]. Subsequent GAN variants address two recurring problems: training instability, handled through gradient-penalty regularization, and constraint compliance, handled through intrusion-detection-specific, function-preserving, or non-functional-feature designs [66], [67], [68], [69]. Other distribution-learning attacks, including attackGAN, NIDS-GAN, quantum generative attacks, and adversarial-retraining pipelines, use learned benign-traffic structure to improve diversity, transferability, or robustness evaluation [70], [71], [72], [73], [74], [75]. However, quantum or quantum-inspired generative attacks should be interpreted as early proof-of-concept studies rather than mature deployable evasion techniques, especially when they rely on simulator-based settings, limited benchmarks, or non-standard datasets.

Beyond GANs, VAE-based methods combine latent generation with active learning, query efficiency, local search, or automatic constraint mining [76], [77]. Autoencoder-based methods preserve temporal coherence or align latent features, while forged-packet and projection-based approaches move generation closer to packet or feature-sequence manipulation [78], [79], [80], [81]. Diffusion-based methods such as TDTCE and NetDiffuser further expand this family by generating diverse traffic through

transformer-based or semantically constrained diffusion processes [82], [83].

Takeaway. Generative attacks shift evasion from direct perturbation toward distributional modeling of benign-like feature and flow representations, with a smaller subset moving closer to packet-level or traffic-level generation. Their strength is diversity and stealth, but their risk is confusing statistical resemblance with operational realism. A benign-looking feature vector or flow record is not necessarily controllable, executable, or malicious-functionality preserving.

F. REINFORCEMENT-LEARNING APPROACHES

Reinforcement-learning approaches treat adversarial evasion as sequential decision-making rather than one-step perturbation. An attack agent modifies traffic through repeated interaction with a detector and uses feedback to choose the next mutation. This is useful when evasion requires dependent changes to flow features, packet properties, or traffic behavior.

The literature differs mainly in where the learned policy operates. At the flow level, [84] uses DQN and SARSA agents to learn mutation policies from binary detector feedback in an offline black-box setting. Its main contribution is the analysis of how learned policies interact with detector feature semantics and functionality-preserving constraints. Other work moves toward live or packet-level adaptation: SE-DQN uses feature-state prediction with atomic packet mutations, Multi-DDQN learns CAN-bus evasion policies that preserve DoS and fuzzing functionality, and NetMasquerade combines Traffic-BERT-based traffic representation with reinforcement learning to rewrite malicious packet sequences into benign-like traffic under hard-label feedback [85], [86], [87]. Across these studies, reinforcement learning enables detector-aware evasion without gradients, but depends strongly on feedback quality, reward design, action-space definition, and the cost of repeated interaction with the detector.

Takeaway. Reinforcement-learning attacks differ because the perturbation is a learned policy rather than a fixed update rule. This supports adaptive, stepwise evasion, but it also makes the attack expensive to learn. The cost of adaptivity is interaction: the policy must learn from feedback, so practicality depends on whether useful behavior emerges before querying becomes too expensive, slow, or detectable.

G. GRADIENT-BASED METHODS

Gradient-based methods adapt adversarial optimization to ML-based NIDS by using model gradients to identify input changes that induce misclassification. They are computationally efficient because they follow the loss gradient rather than exploring many candidate perturbations. In network-security settings, however, gradient updates must be constrained so that modified features remain plausible, semantically meaningful, and protocol-consistent.

C-IFGSM applies iterative gradient updates and then enforces feasibility through clipping and projection [88].

SIGSM embeds validity earlier by masking the update so that only permissible features are modified, particularly in industrial-control-system traffic [89]. The contrast is between correcting invalid gradient movement after the update and restricting the movement before it occurs. Both methods are efficient, but both depend on gradient access and accurate assumptions about which features can be safely changed.

Takeaway. Gradient-based attacks are the most direct way to probe detector sensitivity along the loss surface. Their efficiency is valuable, but it depends on assumptions that are fragile in ML-based NIDS, especially useful gradient access and a faithful constraint model for permissible feature changes. They provide sharp diagnostic pressure on a detector, but their operational value depends on whether gradient-derived edits correspond to valid network behavior.

Overall, attack generation research shows a clear methodological shift: gradient and surrogate methods expose sensitivity under stronger modeling assumptions, search and generative methods relax some access requirements, and domain-constrained or policy-based approaches push toward executable behavior. This trajectory sets up the next question: whether evaluation studies can measure these attacks under conditions that reflect their claimed realism.

H. ADVERSARIAL GENERATION: CRITICAL DISCUSSION

While adversarial generation research in ML-based NIDS has progressed toward attacker-knowledge- and feasibility-aware methods, this should not be read as a movement toward uniformly stronger attacks. The literature instead reveals a recurring trade-off: attacks that are easiest to optimize often rely on idealized access or abstract feature manipulation, whereas attacks that better preserve network semantics, malicious functionality, or limited-knowledge operation usually become more costly, narrower, or harder to transfer.

The main generation-specific tension is the gap between evasion success and operational exploitability. High attack success rate in feature space can expose model sensitivity, but it does not by itself establish that the attack can be executed on a real network. As perturbations become more constrained by protocol validity, functionality preservation, and replayability, the feasible attack space narrows and reported success often declines [90]. This does not make feature-space attacks useless; rather, it means they should be interpreted as diagnostic probes or upper-bound vulnerability tests unless the study demonstrates how the perturbation can be instantiated as valid traffic.

A more specific weakness in attack-generation studies is the underreported actuation layer: the mechanism that converts feature-level changes into concrete packet- or flow-level actions [91]. Even when studies claim practical realism, many do not clearly document how modified features map to timing shifts, header changes, payload edits, packet insertion, traffic shaping, or replayable traces. Without this mapping, the attack cannot be independently reproduced as network

TABLE 7. Analytical summary of adversarial attack-generation families in ML-based NIDS.

Attack family	Core mechanism	Main strengths	Main limitations	Representative studies
Black-box and transferable attacks	Approximate the target detector through surrogate models, limited feedback, manifold learning, or reusable perturbations, then transfer evasive samples to the victim system.	Better match external-adversary settings than white-box attacks and reduce direct dependence on target-model access.	Success remains tied to surrogate fidelity, feature-pipeline alignment, feedback availability, and query budget; transferability may collapse when deployment conditions change.	[26], [27], [29], [35]
Domain-constrained and realistic attacks	Embed protocol rules, feature dependencies, traffic semantics, timing constraints, or problem-space transformations into adversarial generation.	Make evasion claims more operationally meaningful by testing whether perturbations remain valid, functional, and executable as network behavior.	Realism is often gained by narrowing the attack space; constraints are frequently dataset-, protocol-, or pipeline-specific, limiting transfer across environments.	[37], [39], [49], [51]
Evolutionary and heuristic algorithms	Search for evasive perturbations through gradient-free optimization, including genetic algorithms, particle swarm optimization, and fitness-guided traffic mutation.	Suitable for discrete, non-differentiable, and black-box traffic spaces; objectives can combine evasion, edit cost, validity, and interpretability.	Sensitive to fitness design, parameter tuning, and search budget; stronger search can improve evasion but increases query cost and reduces scalability.	[53], [54], [56], [58]
Explainability-guided attacks	Use attribution or saliency signals, such as IG, SHAP, or related XAI methods, to identify decision-critical features and direct perturbations toward them.	Reduce unnecessary feature changes and support more targeted, interpretable, and semantically directed evasion attempts.	Effectiveness depends on attribution stability and fidelity; rankings may vary across models, explanation methods, datasets, and traffic representations.	[59]–[61], [63]
Generative-model attacks	Learn benign-like or less detectable feature/flow distributions using GANs, VAEs, autoencoders, diffusion models, or related generators, then synthesize evasive feature vectors, flows, or perturbations.	Support diversity, stealth, and distributional mimicry, and can generate adversarial examples without direct gradient access to the victim detector.	May confuse statistical resemblance with deployable traffic realism; many remain computationally expensive, unstable to train, surrogate-dependent, feature-space-bound, or proof-of-concept when based on emerging quantum or simulator-based generative methods.	[64], [65], [70], [71], [82], [83]
Reinforcement-learning approaches	Model evasion as sequential decision-making, where an agent learns traffic-mutation policies from detector feedback under functionality or operational constraints.	Capture adaptive, stepwise manipulation and can learn detector-aware policies without gradients.	Require substantial interaction, careful reward and action-space design, and often suffer from high sample or query complexity in dynamic networks.	[84]–[87]
Gradient-based methods	Use model gradients to move malicious samples across the decision boundary, often with clipping, projection, masking, or constraint-aware updates.	Computationally efficient and useful as sharp diagnostic stress tests of detector sensitivity under controlled assumptions.	Usually require white-box access and often remain in feature space; operational value depends on whether gradient-derived edits correspond to valid network behavior.	[88], [89]

behavior, and its practical meaning remains uncertain. For generation research, reporting the permitted actuator set, the feature-to-traffic transformation path, and the validation or replay procedure is therefore as important as reporting the final evasion rate.

Threat-model and cost transparency also determine whether an attack-generation result is practically meaningful. Many attacks achieve strong results under full model access, high-fidelity surrogates, rich feedback, or generous query budgets, while realistic adversaries often face label-only feedback, limited interaction, detection risk, and changing traffic conditions [18]. Query-efficient black-box and reinforcement-learning attacks move closer to this setting, but they often trade success for interaction cost. Generation studies should therefore report the attacker feedback channel, query budget, surrogate or agent training cost, runtime per sample, and the feature-to-traffic actuation path. The broader reporting standard for stealth and attacker-side cost metrics is discussed in Section IX-B.

Overall, adversarial generation should be judged by what each study actually measures. Some attacks measure model sensitivity under idealized access; others measure transferability, constraint-aware evasion, or operational exploitability. Treating all of these as equivalent evidence of real-world adversarial risk obscures the contribution of each method. A stronger generation literature would explicitly

distinguish diagnostic attacks from deployable attacks, report the assumptions that make evasion possible, and show how feature-level perturbations become reproducible network-level behavior.

VII. GENERIC ATTACKS EVALUATION

This section reviews studies that empirically evaluate the robustness of ML-based NIDS against existing adversarial attacks. Unlike attack-generation studies, these works do not primarily introduce new evasion methods; instead, they examine how detectors behave under adversarial conditions, how vulnerability varies across datasets and architectures, and how attack effectiveness transfers between models.

The reviewed studies are organized by evaluation perspective: comparative benchmarking and transferability analysis, domain- and architecture-specific robustness assessment, and realism-constrained viability evaluation. This organization captures a shift from measuring attack success in abstract feature space toward asking whether robustness findings remain meaningful under model, dataset, and network-domain constraints.

A. EFFECTIVENESS, BENCHMARKING & TRANSFERABILITY

Benchmarking studies evaluate how different ML-based NIDS models degrade under adversarial attack and whether robustness claims hold across classifiers, datasets, and

perturbation strategies. Their main contribution is comparative evidence: instead of showing that one attack can fool one detector, they test vulnerability patterns across broader model and attack combinations.

A consistent finding is that robustness is strongly model- and setting-dependent. Comprehensive benchmarks evaluate multiple classifier families under diverse attacks and show that no single detector type is uniformly robust [92], [93], [94], [95], [96], [97], [98]. Search-based and structured perturbation methods, including GA, VAM, and JSMA-focused evaluations, often produce substantial degradation, although stronger attack performance may come with higher computational cost or narrower assumptions [98], [99], [100]. More recent work strengthens benchmarking practice by comparing surrogate- and query-based black-box attacks across heterogeneous classifiers, using statistical robustness measures such as effect sizes and confidence intervals, and showing that both deep and classical detectors can suffer large adversarial performance drops despite strong clean-traffic accuracy [101], [102], [103], [104], [105].

Transferability studies examine whether attacks crafted for one model remain effective against another. Their results show that transfer success depends less on attack strength alone than on surrogate–target alignment, feature representation, and architectural similarity [106]. Related work extends this concern to training-time and sequence-model vulnerabilities, including poisoning and classifier-poisoning attacks that degrade detection by altering training data or internal model dynamics [107], [108]. Other evaluations show that A2PM-style feature-constrained generators, generative attacks, deployment-phase perturbations, and feature-specific budgets produce different transferability patterns depending on the model family and perturbation assumptions [109], [110], [111], [112], [113]. Even simple gradient-based attacks can substantially degrade performance, but transferability claims are most informative when reported alongside the feature representation, surrogate–target relationship, and perturbation assumptions used in the evaluation [114], [115], [116].

Takeaway. Benchmarking and transferability studies show that adversarial robustness in ML-based NIDS is not a stable property of a classifier family; it depends on the interaction among attack type, preprocessing, feature representation, surrogate–target similarity, and evaluation metric. Their value lies in exposing comparative vulnerability patterns, but their results remain difficult to generalize when datasets, attack settings, and reporting protocols differ. The evidence supports standardized evaluation, while the broader implications of this heterogeneity are addressed later in the discussion.

B. DOMAIN AND ARCHITECTURE-SPECIFIC ANALYSES

Domain- and architecture-specific evaluations ask whether adversarial robustness changes across network environments, traffic representations, and detector designs. Rather than treating ML-based NIDS as a uniform target, these studies

examine how deployment context and model architecture shape vulnerability.

Comparisons between traditional machine learning and deep learning show that clean-data performance does not reliably predict adversarial resilience. Encrypted-traffic, general NIDS, and IoT evaluations report that robustness varies with attack type, feature representation, and model design rather than following a simple traditional-versus-deep-learning divide [117], [118], [119]. Architecture-focused studies reach a similar conclusion: shallower networks, reduced feature sets, suitable activations, model depth, and input structure can all affect robustness, but none provides universal protection [120], [121], [122].

Specialized detection settings reveal more specific vulnerability profiles. In software-defined networking, attack impact depends on both the detector architecture and the perturbation phase [123]. Reinforcement-learning-based detectors are also vulnerable, with robustness shaped by the specific control algorithm, internal state representation, and targeted perturbation strategy [124], [125], [126]. Other evaluations examine self-normalizing networks, GAN-assisted classifiers, IoT deep architectures, capsule-based detectors, hybrid convolutional–recurrent–transformer models, and explanation-based feature-manipulation analysis, showing that specialized architectures may shift the form of vulnerability rather than eliminate it [127], [128], [129], [130], [131], [132].

Takeaway. Domain- and architecture-specific evaluations show that clean-data performance and architectural choice are weak predictors of adversarial resilience. Their main contribution is to weaken broad claims that any single architecture family is inherently robust. Architecture can improve resilience in particular settings, but the evidence does not support universal conclusions across NIDS environments [5], [7].

C. REALISM-CONSTRAINED VIABILITY

Realism-constrained viability studies evaluate whether adversarial examples that succeed in feature space remain feasible under network-domain constraints. Their focus is not only whether an attack fools a detector, but whether the modified traffic can remain valid, functional, and consistent with the relationships present in real network data.

A central contribution of this literature is the development of quantitative measures for attack realism. Feasibility and unrealism metrics evaluate whether transfer success depends on unrealistic feature alignment, invalid perturbations, or broken traffic relationships [90], [133]. Protocol-aware validation pipelines and feature-relationship consistency analyses extend this direction by checking whether adversarial examples preserve domain constraints beyond simple numerical similarity [134], [135]. These studies make evaluation more rigorous by distinguishing high evasion rates from practically meaningful vulnerability.

A complementary set of studies compares constrained and unconstrained evaluations. They show that enforcing

value ranges, semantic consistency, and feature dependencies often lowers attack success and transferability, although the magnitude varies by attack method and constraint strength [136], [137]. This finding is important because a lower success rate under constraints may represent a more valid evaluation rather than a weaker attack.

Takeaway. Realism-constrained evaluation changes the interpretation of adversarial success: an attack is not fully informative if it only succeeds by producing invalid or non-functional traffic. These studies contribute metrics and validation procedures that separate numerical evasion from operational feasibility. Their limitation is scope: current realism checks capture only selected constraints, so they improve validity without yet providing a complete model of network behavior [6], [18].

D. EVALUATION PARADIGMS: CRITICAL DISCUSSION

Adversarial evaluation research in ML-based NIDS has become more systematic, but its central weakness is not simply the choice of metric or attack. The deeper issue is that many evaluations measure robustness within highly shared experimental pipelines: the same datasets, feature extraction routines, preprocessing choices, and classifier families are often reused across studies. As a result, reported robustness and transferability may reflect the structure of the evaluation setup as much as the intrinsic resilience of the detector.

This problem is most visible in transferability evaluation. Transfer attacks are intended to test whether adversarial examples crafted against one model remain effective against another, but many studies evaluate transfer across models trained on the same dataset and represented through the same feature pipeline [63]. Under these conditions, high transfer success may indicate shared feature dependencies or preprocessing artifacts rather than genuine architecture-independent vulnerability. This is especially problematic when evaluations rely on legacy datasets or narrow threat-model assumptions, because the resulting transferability claims may not generalize to different traffic distributions, feature schemas, or deployment environments [18], [95].

Realism-aware metrics such as feasibility scores, Transferability Feasibility Score, and Unrealism Index have started to expose this limitation by showing that high attack success can collapse when validity, executability, or feature-relationship constraints are enforced [90], [107], [136]. For evaluation studies, their value is not merely that they penalize unrealistic attacks, but that they reveal when a robustness conclusion depends on an overly permissive experimental setup. In this sense, realism metrics should be treated as diagnostic checks on the evaluation protocol, not only as additional attack-performance indicators.

A stronger evaluation paradigm should therefore separate three claims that are often conflated: *whether an attack reduces detector performance*, *whether it transfers across genuinely different pipelines*, and *whether it remains operationally feasible*. These questions require different evidence. Performance

degradation can be tested within a benchmark; transferability requires variation in datasets, feature extractors, preprocessing routines, and architectures; and operational relevance requires validity and cost reporting. When these dimensions are collapsed into a single attack success rate, evaluation studies risk overstating both adversarial vulnerability and defensive robustness.

The main implication is that adversarial evaluation should be designed to test generalization, not only degradation. Robustness claims become more credible when models are evaluated across heterogeneous datasets, independent feature pipelines, varied threat models, and transparent reporting conditions [6]. Without this separation, evaluation results remain difficult to compare and may describe robustness within a benchmark ecosystem rather than resilience in deployed ML-based NIDS.

VIII. DEFENSE MECHANISMS

This section reviews studies that propose and evaluate defense mechanisms for ML-based NIDS against adversarial evasion. These works aim to improve detection reliability by strengthening model training, filtering suspicious inputs, transforming perturbed samples, embedding robustness into architectures, or coordinating multiple protective components. Unlike evaluation studies that primarily expose vulnerabilities, defense studies seek to reduce the practical impact of evasion while preserving acceptable performance on legitimate traffic.

The reviewed defenses are organized according to where robustness is introduced in the detection pipeline: during training; before classification through detection or filtering; through interpretability or certification; through input purification; within the model architecture; or through coordinated multi-stage protection. This organization highlights the field's movement from isolated countermeasures toward layered robustness strategies that combine learning, detection, preprocessing, and architectural design.

A. ADVERSARIAL TRAINING-BASED DEFENSES

Adversarial training-based defenses harden ML-based NIDS by exposing models to adversarial examples during training. The core idea is simple: perturbations that would otherwise maximize model loss are incorporated into learning so that the detector becomes less sensitive to similar evasive inputs at test time. In practice, this family differs mainly in how adversarial examples are generated, mixed with clean data, and used to reshape decision boundaries.

Standard adversarial training uses gradient-based attacks such as FGSM or PGD to generate training perturbations. Min-max training with iterative PGD, exclusive or mixed adversarial-sample schedules, and neural intrusion-detector retraining all show that adversarial exposure can recover robustness against evaluated white-box attacks [138], [139], [140]. However, the gains are not uniform: some studies report broad improvements across architectures, while others show that stronger perturbations or different attack settings can still expose residual vulnerability [141], [142].

Enhanced variants try to widen the robustness gained from retraining. GAN-informed frameworks such as AIDTF and generative augmentation methods synthesize stronger or more diverse adversarial examples [143], [144]. Sanitization-driven retraining combines leakage-safe preprocessing, balancing, generated attack synthesis, and adversarial retraining, while LARAR targets vulnerable hidden layers through adaptive regularization and auxiliary supervision [145], [146]. Hybrid adversarial retraining broadens exposure by combining white-box and black-box variants, whereas sensitivity-based training reduces feature sensitivity directly [147], [148]. These methods improve robustness coverage, but also increase training cost and remain sensitive to the attacks used during retraining.

Takeaway. Adversarial training is the most direct defense against evasion because it teaches the detector using adversarially perturbed inputs rather than only clean traffic. Its strength is threat-model specificity: it can be highly effective when training attacks resemble evaluation attacks. That same specificity is its weakness, since robustness may not transfer to unseen, adaptive, or black-box perturbations, and stronger training often competes with clean accuracy and computational efficiency [5], [19].

B. DETECTION & FILTERING OF ADVERSARIAL SAMPLES

Detection and filtering defenses add an auxiliary screening layer before or alongside the primary classifier. Instead of modifying the detector itself, they attempt to identify adversarial inputs through reconstruction errors, activation behavior, distance scores, constraint violations, or combinations of these signals. Their appeal is modularity: in many cases, they can be attached to existing NIDS without rebuilding the full classifier.

Single-signal detectors rely on one dominant source of evidence. READ, NIDS-DA, and AE-Gate use reconstruction error or autoencoder-based gating to detect samples that deviate from learned normal relationships [149], [150], [151]. Activation-based detection trains a separate classifier on internal neuron activations, while DDM-CNN uses distance-weighted density for out-of-distribution detection [152], [153]. These methods are relatively interpretable and modular, but their reliability depends on whether adversarial perturbations consistently disturb the chosen signal.

Multi-signal frameworks attempt to reduce this dependence by combining complementary evidence. MANDA fuses manifold consistency with decision-boundary proximity, while AdvAT strengthens boundary-aware detection through iterative perturbation [154], [155]. NIDS-DEFEND combines batch-level testing with individual classification, FGMD uses feature grouping and model fusion, and NIDS-CBAD detects violations of domain-specific feature relationships without requiring adversarial examples for training [156], [157], [158]. AAD extends this compositional direction through request validation, adversarial discrimination, and a hardened ensemble IDS [159]. Detect & Reject represents a different design choice by explicitly retraining with adversarial

examples to create a reject class [160]. These systems improve coverage across perturbation types, but add calibration, latency, and adaptive-attack concerns.

Takeaway. Detection and filtering defenses are strongest when adversarial inputs leave detectable traces in reconstruction behavior, activation patterns, distributional distance, or domain constraints. Their modularity makes them practical, but also exposes their central fragility: once the detector signal is known, an adaptive adversary can optimize against both the classifier and the filter. The main design problem is therefore not only detecting perturbations, but keeping the detection signal stable under shifting traffic and detector-aware attacks [5].

C. EXPLAINABILITY & FORMAL VERIFICATION

Explainability and formal verification defenses address robustness from two different angles: interpretability seeks to expose why a detector behaves suspiciously, while verification seeks to prove that predictions remain stable within a defined perturbation region. Both move beyond raw accuracy, but they provide different kinds of assurance.

Explainability-driven defenses use attribution or visualization signals to reveal adversarial behavior. NIDS-Vis combines visualization, feature-space partitioning, and distributional regularization to encourage group-wise invariance [161]. Other studies combine adversarial training with SHAP explanations, use LIME explanation mismatches as adversarial signals, or model SHAP attribution vectors as behavioral fingerprints [162], [163], [164]. These methods make model behavior more inspectable and can reveal feature-level manipulation, but their defensive value depends on explanation fidelity.

Formal verification offers a stronger but narrower form of assurance. Interval Bound Propagation certifies that predictions remain unchanged within a perturbation bound around each input, providing per-input robustness guarantees rather than empirical evidence alone [165]. The trade-off is scalability: certification becomes difficult for large, mixed-feature, or protocol-constrained NIDS models, and common norm-bounded assumptions may not capture realistic network manipulations.

Takeaway. Explainability and verification improve trust in different ways: explanation makes adversarial behavior inspectable, while verification makes bounded robustness certifiable. The former is more flexible but not a proof; the latter is stronger but harder to scale and often tied to simplified perturbation models. Their shared challenge is adapting transparency and certification to discrete, dependent, and protocol-constrained network features [3].

D. INPUT PURIFICATION & FEATURE-SPACE DEFENSES

Input purification defenses attempt to remove or neutralize adversarial perturbations before classification. They operate as preprocessing modules that project suspicious inputs back toward a clean or expected data manifold, making them

attractive when the underlying NIDS cannot be redesigned. Their effectiveness depends on whether adversarial changes behave like removable noise rather than detection-relevant signal.

Reconstruction-based purification is the dominant design. Denoising autoencoders, RePO/RePO+, and autoencoder-denoising pipelines reconstruct perturbed inputs before classification, aiming to recover cleaner feature representations [166], [167], [168]. BiGAN combines reconstruction error with discriminator realism scores, adding a generative consistency signal to adversarial detection [169]. These approaches are modular and model-agnostic, but may introduce latency or alter benign and malicious features in unintended ways.

More complex purification methods use adversarial or diffusion-based generation. APE_GAN++ filters perturbations with a GAN-based preprocessor, while diffusion-based purification iteratively projects inputs toward the learned data manifold [170], [171]. These methods can handle diverse perturbation patterns, but their iterative or generative nature increases inference cost and may be unsuitable for real-time deployment.

Takeaway. Input purification is appealing because it can be inserted before an existing detector, but it rests on a delicate assumption: the adversarial component can be removed without erasing malicious evidence or distorting benign traffic. Reconstruction and generative purification can improve robustness, yet they also create new failure modes through latency, over-smoothing, and adaptive attacks that account for the purifier. The question for this family is not only whether inputs can be denoised, but whether purification preserves the security meaning of the traffic [19].

E. ROBUST MODEL ARCHITECTURES & ENSEMBLES

Robust architectures and ensembles embed resilience into the detector itself. Instead of adding a separate filter or purifier, these defenses use diversity, hierarchy, structural modeling, or specialized learning paradigms to make evasion harder. Their common goal is to force the attacker to bypass multiple decision views or exploit more complex model structure.

Ensemble and hierarchical designs use diversity as the main source of robustness. AppCon uses application-aware partitioning with per-application classifiers, RSE combines stacking with autoencoder-based detection and denoising, and random feature scheduling increases attacker uncertainty [172], [173], [174]. Hybrid robust IDS frameworks combine adversarial-aware learning, sanitization, anomaly filtering, feature squeezing, and layered defenses to recover performance under adversarial manipulation [175], [176]. Hierarchical Learning and hierarchical adversarially robust IDS designs organize detection into multi-level decisions, while RAIDS and Context-Aware IDS fuse reconstruction, ensemble outputs, distributed decision-making, or denoising modules [177], [178], [179], [180]. AFP takes a different route

by disrupting the attacker's black-box feedback loop through traffic profiling and selective feature perturbation [181].

Other work builds robustness through specialized model structure. Graph neural networks and optimized graph-neural IDS frameworks exploit host-connection structure, while quantum-kernel, resource-efficient, edge-oriented, and constraint-aware tabular designs explore robustness under different deployment constraints [182], [183], [184], [185], [186], [187]. Random-Forest Distillation smooths decision boundaries, ARGAN-IDS neutralizes inputs through GAN-based reconstruction, semi-supervised specialization exploits cluster structure, and RNN-ADV uses architectural selection to improve resilience [188], [189], [190], [191]. These approaches show that architecture can contribute to robustness, but often at the cost of implementation complexity, maintenance overhead, or resource demand.

Takeaway. Robust architectures and ensembles make evasion harder by increasing decision diversity or embedding structural assumptions into the detector. Their strength is that robustness is partly built into the model rather than attached as an external module. Their limitation is operational: more components, hierarchies, or specialized structures can improve resilience in experiments while making the system harder to deploy, maintain, and adapt across environments [5], [7].

F. HYBRID & MULTI-STAGE FRAMEWORKS

Hybrid and multi-stage frameworks combine several defense principles into coordinated pipelines. They reflect a defense-in-depth strategy: rather than relying on one mechanism, they distribute protection across adversarial training, feature processing, ensemble diversity, dynamic selection, purification, detection, and fallback classification.

One group integrates adversarial training with complementary robustness components. ELAT combines ensemble learning with adversarial training, RobEns adds feature squeezing and ensemble methods, and Def-IDS uses GAN-based augmentation with multi-source adversarial retraining [192], [193], [194]. Other frameworks combine adversarial training with balancing, feature engineering, spatial smoothing, optimization, SHAP-guided feature selection, or GAN-based filtering [195], [196], [197], [198]. These combinations widen defensive coverage, but also increase inter-component dependence.

A second group emphasizes dynamic selection or staged processing. Apollon, DYNAMITE, and IDS-Anta select classifiers or defenses using bandits, clustering, prediction models, or ant-colony optimization [199], [200], [201]. DLL-IDS and MSAAD combine adversarial detection, fallback classification, purification, or ensemble selection, while fixed two-stage designs use noise filtering, feature extraction, principal-component analysis, or GAN-based training [202], [203], [204], [205]. These systems improve adaptability and unpredictability, but their effectiveness depends on whether the stages coordinate reliably under real-time constraints.

Takeaway. Hybrid and multi-stage frameworks are attractive because they do not bet robustness on a single mechanism. Their layered design can cover more attack paths, but it also expands the attack surface and increases latency, calibration burden, and maintenance complexity. The value of this family depends on whether combined defenses produce genuine system-level robustness rather than a collection of individually promising modules [18].

G. DEFENSE MECHANISMS: CRITICAL DISCUSSION

Adversarial defense research in ML-based NIDS has moved from isolated countermeasures toward layered systems that combine training, detection, purification, architectural diversity, and dynamic selection. This shift is important, but it should not be interpreted as evidence that current defenses have achieved general robustness. The stronger conclusion is more cautious: most defenses are conditionally robust, succeeding when their assumptions about the attack, model architecture, feature representation, and deployment setting remain aligned.

This conditionality appears across defense families. Adversarial training improves resilience when the attacks used during training resemble those used during evaluation, but its protection can weaken under unseen, black-box, or functionality-preserving attacks [142]. Detection and filtering methods are effective when adversarial perturbations leave stable anomaly cues, yet adaptive adversaries can target the detector signal itself [154]. Purification methods assume that adversarial changes can be removed without erasing malicious evidence, an assumption that becomes fragile when perturbations are crafted to survive denoising. Robust architectures, ensembles, and hybrid frameworks increase diversity and attacker effort, but they may still share preprocessing assumptions, feature dependencies, or component-level weaknesses [192], [193]. Thus, the central weakness is not that individual defenses fail, but that their robustness often remains tied to narrow threat–architecture pairings and is sometimes purchased at the cost of clean-traffic accuracy.

A second issue is that layered defense does not automatically imply system-level robustness. Combining multiple components can widen coverage, but it can also introduce coordination failures, redundant modules, calibration burden, latency, and new attack surfaces. A hybrid pipeline is only as robust as the interaction among its parts: a purifier that distorts security-relevant features, a detector calibrated to static perturbations, or an ensemble sharing the same feature extractor can all undermine the apparent benefit of layering. This means that defense studies should evaluate not only whether each component improves adversarial accuracy, but whether the complete pipeline remains stable under changing traffic, unseen attacks, and realistic operational constraints.

The main critical distinction is therefore between *component robustness* and *deployable resilience*. Component robustness shows that a defense improves performance

under a selected attack setting. Deployable resilience requires stronger evidence: robustness across different attack families, architectures, feature pipelines, datasets, and adaptive conditions. Many current defenses demonstrate the first form convincingly but provide limited evidence for the second [5], [6], [7]. For this reason, defense research should avoid treating improved adversarial accuracy under a fixed benchmark as proof of general protection.

Overall, the defense literature shows that robustness in ML-based NIDS is not a property of a single module but an emergent property of the full detection pipeline. A high-quality defense claim should make explicit what assumptions the defense depends on, which attacks and feature representations it covers, and whether robustness persists when the surrounding pipeline changes. Without this distinction, defenses risk being reported as broadly protective when they are in fact effective only within a narrow experimental envelope.

IX. DISCUSSION

The critical discussions in Sections VI, VII, and VIII diagnose limitations *within* each domain: how evasive examples are generated, how robustness is measured, and how defenses are built and validated. This section forms the second tier of the analysis and does not restate those domain-level findings. It examines instead the three issues that become visible only when the domains are read against one another: how the perturbation-constraint regime determines what a result *means* (Section IX-A), how reporting practice determines whether a result can be *audited* (Section IX-B), and how empirical and operational conditions bound how far a result *generalizes* (Section IX-C).

A. PERTURBATION CONSTRAINTS AND REALISM LEVELS

Across adversarial NIDS research, perturbation constraints define which changes to flow- or packet-level representations are admissible during attack crafting, evaluation, or defense testing. These constraints determine whether adversarial examples are merely numerical distortions of feature vectors or traffic instances that could plausibly exist on a live network. A cross-strand pattern is clear: generation studies more often encode domain or protocol restrictions to preserve executability, whereas evaluation and defense studies frequently reuse generic attacks such as FGSM, PGD, or CW directly in NIDS feature spaces without enforcing network-specific semantics.

1) NO CONSTRAINTS

In the weakest setting, adversarial examples are produced by applying generic feature-space attacks without restrictions on protocol legality, temporal consistency, feature dependencies, or malicious functionality [97], [118], [119]. The adversary is therefore implicitly allowed to rewrite feature vectors regardless of whether the resulting points correspond to realizable traffic. Robustness or defense effectiveness measured under this regime should be interpreted cautiously,

TABLE 8. Analytical summary of adversarial defense mechanisms in ML-based NIDS.

Defense family	Core mechanism	Main strengths	Main limitations	Representative studies
Adversarial training-based defenses	Retrain the detector using adversarially perturbed samples, often through min–max learning, GAN-informed augmentation, sensitivity reduction, or layer-wise regularization.	Directly harden classifiers against known evasion patterns and can recover detection performance when training and evaluation attacks are aligned.	Robustness is threat-model specific; performance may degrade under unseen, adaptive, black-box, or functionality-preserving attacks, and stronger training can increase cost or reduce clean-data accuracy.	[138], [139], [143], [146]
Detection and filtering of adversarial samples	Add auxiliary screening layers that detect adversarial inputs through reconstruction error, activation behavior, distance scores, manifold consistency, boundary proximity, or feature-constraint violations.	Modular and often attachable to existing NIDS without rebuilding the primary classifier; multi-signal designs improve coverage across perturbation types.	Reliability depends on detection-signal stability; adaptive adversaries can optimize against both classifier and filter, while calibration and latency remain concerns.	[149], [150], [154], [156]
Explainability and formal verification defenses	Use explanation signals to expose suspicious attribution behavior, or formal methods to certify that predictions remain unchanged within a bounded perturbation region.	Move robustness assessment beyond raw accuracy; explainability improves inspectability, while verification provides stronger per-input robustness guarantees.	Explanation-based defenses depend on attribution fidelity and stability, whereas formal verification is difficult to scale to mixed-feature and protocol-constrained NIDS settings.	[161], [162], [164], [165]
Input purification and feature-space defenses	Preprocess traffic representations before classification by reconstructing, denoising, or projecting perturbed inputs toward a clean or expected data manifold.	Model-agnostic and insertable before existing detectors; reconstruction and generative purification can neutralize diverse perturbation patterns.	Assume adversarial noise can be removed without erasing malicious evidence; purification can introduce latency, over-smoothing, clean-traffic degradation, and vulnerability to adaptive attacks.	[166], [167], [169], [170]
Robust architectures and ensembles	Embed resilience into the detector through diversity, hierarchy, graph structure, application-aware partitioning, quantum or edge-oriented designs, distillation, or specialized robust learning.	Increase evasion difficulty by forcing attackers to bypass multiple decision views or exploit more complex structural assumptions.	Architectural complexity can increase computation, maintenance burden, and deployment difficulty; ensembles may still share preprocessing assumptions or correlated weaknesses.	[181]
Hybrid and multi-stage frameworks	Combine multiple defenses, such as adversarial training, purification, detection, feature engineering, ensemble learning, and dynamic model or defense selection, into coordinated pipelines.	Offer defense-in-depth and adaptability by allowing complementary components to cover different attack stages, perturbation types, or deployment conditions.	Layering does not guarantee system-level robustness; multi-stage pipelines can suffer from coordination failures, redundant modules, latency, and new attack surfaces.	[192]–[194], [199], [200], [203]

because the attack may exploit invalid regions of the feature space rather than expose an implementable network threat [90], [136].

2) MATHEMATICAL MAGNITUDE CONSTRAINTS

Norm-bounded perturbations restrict the numerical size of a change through L_∞ , L_2 , or more general L_p bounds. This provides a clean model-centric measure of perturbation magnitude and underpins common gradient-based methods such as FGSM, BIM, PGD, and CW [91]. However, a small numerical change is not necessarily a meaningful or valid network change. Such constraints control how much a vector changes, but not whether the modified features preserve protocol semantics, traffic dependencies, or executable behavior.

3) FEATURE-SPACE ABSTRACTION CONSTRAINTS

Feature-space constraints arise when perturbations are restricted to preprocessed tabular or flow-level features rather than raw network traffic. This setting constrains where the attack operates, but it does not guarantee that the modified feature vector can be mapped back to a concrete packet trace [39]. As a result, attacks, evaluations, or defenses developed under this assumption may measure sensitivity to synthetic feature distortions rather than robustness against deployable adversarial flows. This limitation also affects purification defenses: if a sanitized or reconstructed sample is not checked against network semantics, improved classification after preprocessing may not imply recovery of valid traffic behavior.

4) PROTOCOL SEMANTICS CONSTRAINTS

Protocol semantics constraints embed domain rules into the perturbation process, such as valid field ranges, TCP/IP header coherence, legal flag combinations, IoT flow policies, or consistent sequence-number behavior [137]. These constraints substantially improve low-level realism because they restrict perturbations to syntactically feasible packet or flow attributes. Nevertheless, protocol compliance alone is insufficient: a trace may be syntactically valid while no longer preserving the original malicious objective.

5) MALICIOUS FUNCTIONALITY PRESERVATION CONSTRAINTS

Functionality-preservation constraints require that the perturbed traffic continue to implement the underlying attack behavior and retain its ground-truth malicious label. For example, a perturbed denial-of-service flow, payload delivery, command-and-control exchange, or session behavior must still achieve its malicious purpose while evading detection. Problem-space generators explicitly address this requirement by enforcing behavioral or label preservation in addition to domain validity [39], [49], [84]. This makes functionality preservation a stronger constraint than protocol compliance, because it requires the adversarial example to remain not only valid, but operationally malicious.

6) PACKET-LEVEL EXECUTABILITY CONSTRAINTS

Packet-level executability constraints, the most rigorous level of enforcement, extend perturbation design to concrete traffic traces. Under this regime, an adversarial example must exist as

a valid, injectable packet stream: checksums must be correct, sequence numbers and session state must remain consistent, timing must be plausible, and the resulting traffic should be validated through replay or emulation. This level upgrades feature- or protocol-level validity into end-to-end operational evidence that the crafted sample can be instantiated on a network [39], [48], [49].

Taken together, the literature still relies heavily on unconstrained, norm-bounded, or feature-space perturbations, especially in evaluation and defense studies, while only a smaller subset of generation work enforces protocol semantics, malicious functionality, or packet-level executability. This imbalance matters because high ASR under weak constraint regimes should be treated as an upper-bound vulnerability estimate, not as direct evidence of operational exploitability. The constraint level therefore determines the meaning of a robustness claim: the stronger the admissibility requirements, the closer the result is to adversarial traffic that could actually be implemented in a real NIDS environment.

B. METHODOLOGICAL VALIDITY, REPORTING, AND REPRODUCIBILITY GAPS

Section VIII showed that defense robustness is conditional on the alignment between attack, architecture, and feature pipeline. The concern here is broader and applies equally to generation, evaluation, and defense studies: whether the assumptions behind *any* reported robustness claim are stated fully enough for the claim to be checked. Across adversarial NIDS research, many robustness claims remain difficult to reproduce, compare, or interpret because the underlying methodological assumptions are incompletely specified. These weaknesses are not merely empirical limitations; they affect how adversarial capability, robustness, and defense effectiveness are defined. Four recurring gaps are especially important: threat-model realism, reporting transparency, attack–defense integration, and feature-pipeline dependence.

1) THREAT-MODEL FEASIBILITY AND ATTACKER VANTAGE

Many studies grant attackers stronger capabilities than are plausible in deployed NIDS environments. White-box or full-knowledge assumptions often provide access to model parameters, gradients, or internal feature mappings, while even nominally black-box settings may allow generous query budgets, stable input–output observability, or rich feedback channels [92], [97], [124]. Some works also assume network-level vantage points such as full packet interception, fine-grained timing manipulation, or direct access to feature representations of encrypted traffic [39], [49], [84]. Notably, several of these works are also among the strongest examples of problem-space realism, enforcing protocol validity and malicious-functionality preservation; this highlights that problem-space realism and threat-model realism are distinct properties: an attack can preserve functional, executable behavior while still presupposing a privileged vantage point, so the two should be assessed and reported separately. These

assumptions may be useful for controlled stress testing, but they should be reported as such; otherwise, they can make attacks appear more practical, and defenses appear weaker, than they would be under realistic attacker visibility.

2) REPORTING AND REPRODUCIBILITY STANDARDS

A second methodological gap is the absence of consistent reporting for implementation and feasibility details. Critical elements such as feature scaling, preprocessing pipelines, random seeds, attack configurations, dataset splits, perturbation budgets, and success definitions are often omitted, while evaluation metrics vary across studies from accuracy to precision, recall, F1-score, AUC, attack success rate, or detection-rate degradation. This makes cross-study comparison fragile even when studies appear to evaluate similar attacks or defenses. A field-wide reporting standard should therefore capture attacker-side feasibility, not only detector-side performance. Beyond conventional metrics, adversarial NIDS studies should disclose the threat-model class, attacker feedback channel, query budget, surrogate-training cost, runtime per sample, and compute resources. For attack studies, two additional indicators would improve comparability. The first is a stealth indicator, capturing side effects such as false-positive shifts, timing irregularities, packet-arrival entropy, or other detectable traffic artifacts. The second is an attacker-effort efficiency indicator, which this review proposes as the number of successful evasions achieved per unit of attacker cost, such as per 1,000 model queries, per GPU-hour, or per generated adversarial example. This indicator is not intended as a standardized metric, but as a reporting measure that helps compare the practical cost of black-box, transfer-based, optimization-based, and reinforcement-learning-based attacks.

3) ATTACK–DEFENSE INTEGRATION

Attack and defense studies are often evaluated in isolation, which limits understanding of how adversarial robustness behaves when both sides are considered as interacting components. Generation studies may demonstrate evasion against static detectors without testing whether the attack remains effective against contemporary defense mechanisms; defense studies may rely on outdated or simplified attacks; and evaluation studies rarely benchmark newly proposed defenses within the same experimental framework. This separation obscures the interaction between evasive behavior and defensive response, allowing isolated experiments to overstate both attack persistence and defense reliability. The methodological issue is not simply that experiments are offline, but that attacks and defenses are rarely evaluated as coupled parts of the same security system [5].

4) FEATURE-PIPELINE DEPENDENCE AND ALGORITHMIC FRAGILITY

Robustness findings are also highly sensitive to the feature-processing pipeline used during design and evaluation. Attacks and defenses can perform well on the original dataset,

feature schema, or model family, yet degrade when transferred to new feature extractors, aggregation windows, encoders, normalization rules, architectures, or hyperparameter settings [139], [141]. This dependence can make robustness appear stronger than it is, because the reported result may reflect alignment with a particular preprocessing pipeline rather than resilience to adversarial manipulation. Since systematic cross-feature, cross-architecture, and cross-pipeline evaluations remain limited, hidden dependencies on specific representations often remain untested [149], [156].

Overall, methodological validity in adversarial NIDS research depends on making assumptions auditable. Threat models should describe what the attacker can observe and control; experiments should report enough detail to be reproduced; attacks and defenses should be evaluated as interacting mechanisms; and robustness claims should be tested beyond a single feature pipeline. Without these safeguards, reported improvements may describe performance under a narrow experimental construction rather than reliable adversarial robustness.

C. EMPIRICAL, OPERATIONAL, AND LIFECYCLE CONSTRAINTS

Even when adversarial NIDS studies are methodologically sound, their conclusions often remain bounded by the empirical and operational conditions under which they are tested. These constraints concern the external validity of the evidence: whether the datasets, deployment pipelines, resource assumptions, and temporal evaluation settings reflect the environments in which ML-based NIDS are expected to operate.

1) DATASET NARROWNESS AND BENCHMARK STALENESS

A large portion of the literature still depends on a small set of recurring benchmarks, especially NSL-KDD, CICIDS2017, and UNSW-NB15 [117], [142], [173], [206], [207]. These datasets remain useful for comparison, but they cannot fully represent contemporary traffic diversity, encrypted communication, IoT and cloud workloads, or emerging attack behavior. Evaluation scope is also often narrow: many studies use one or two datasets and a limited attack set, making it difficult to assess whether reported robustness extends to unseen attacks, different traffic distributions, or modern operational contexts [5]. The issue is not simply that older datasets are invalid, but that over-reliance on them can make adversarial robustness appear more general than the empirical evidence supports.

2) PIPELINE-BLIND EVALUATION

Many studies evaluate models on static CSV, flow, or PCAP representations with limited attention to how traffic is processed inside a deployed NIDS pipeline [5]. Real systems involve packet ingestion, flow reconstruction, normalization, sampling, encryption constraints, and intermediate components such as firewalls, proxies, load balancers, and traffic

normalizers. These stages can rewrite, suppress, or reshape both benign traffic and adversarial perturbations before classification. When evaluations bypass this pipeline, they may test model behavior under simplified inputs rather than end-to-end detection behavior under operational traffic handling.

3) SCALABILITY AND SYSTEM-LEVEL PROFILING

Operational viability also depends on whether defenses and detection pipelines can meet real-time requirements. Complex mechanisms such as multi-generator models, diffusion-based purifiers, iterative adversarial retraining, and hybrid multi-stage defenses can impose substantial computational and memory overheads [141], [170], [203]. Yet many studies report adversarial accuracy or robustness improvement without profiling training duration, inference latency, throughput degradation, CPU/GPU demand, memory footprint, or energy consumption under benign and adversarial load [5]. Without such system-level profiling, it remains unclear whether a proposed method can operate within the latency and throughput constraints of real NIDS deployments.

4) LIFECYCLE AND CLOSED-LOOP VALIDATION

A final operational limitation is that most studies evaluate robustness as a single-phase snapshot. Models, thresholds, denoisers, ensemble weights, and feature subsets are typically fixed after training, while traffic distributions and adversarial behavior are treated as stable [5]. In practice, deployed NIDS must cope with concept drift, changing attack frequency, retraining cycles, threshold adaptation, and defense switching. Lifecycle-aware evaluation should therefore examine whether robustness persists over time as the model, traffic, and adversary evolve. This is distinct from merely testing attacks and defenses together: the key question is whether the full detection system remains reliable under repeated updates, shifting traffic, and prolonged adversarial exposure.

5) WORKED EXAMPLE: APPLYING THE PROPOSED INDICATORS

To make the two proposed indicators concrete rather than purely definitional, they are applied here to three representative generation studies from the reviewed corpus. The *composite realism score* (CRS) introduced in Direction (D1) is computed as the mean of three components—protocol validity, malicious-functionality preservation, and replayability or executability—each scored as not enforced (0), asserted but not validated (0.5), or enforced and validated (1). The *attacker-effort efficiency* indicator defined above is reported only where the source study discloses the attacker-side cost required to mount the attack. Table 9 summarizes the result.

The example shows what the score discriminates. IDSGAN restricts perturbations to the non-functional features of NSL-KDD records and never reconstructs packet traffic, so despite a high reported evasion rate it supports only a diagnostic, feature-space claim (CRS 0.33). Adv-Bot enforces syntactic and semantic domain constraints on realistic botnet traces

TABLE 9. Worked example: composite realism score (CRS) for three representative generation studies. PV = protocol validity; FP = functionality preservation; EX = replayability/executability; each scored 0 (not enforced), 0.5 (asserted, not validated), or 1 (enforced and validated); CRS is their mean. Cost = whether attacker-side cost is reported.

Study		PV	FP	EX	CRS	Cost
IDSGAN [67]	(feature-space)	0.5	0.5	0	0.33	No
Adv-Bot [39]	(problem-space)	1	1	0.5	0.83	No
TANTRA [49]	(packet-level)	1	1	1	1.00	No

and preserves the underlying attack logic under black-box knowledge assumptions, but stops short of end-to-end replay validation (CRS 0.83). TANTRA reshapes only inter-packet timing and leaves packet content unchanged, so malicious functionality is preserved by construction and the reshaped traffic remains directly executable (CRS 1.00). All three report high attack success, yet they support markedly different operational claims, which is precisely the distinction the score is intended to expose. The attacker-effort column is equally instructive: none of the three studies reports the query budget, runtime, or compute cost needed to mount the attack, so the efficiency indicator cannot be computed from the published record. This is not a limitation of the indicator but direct evidence for the auditable reporting standard called for in Direction (D4).

Overall, empirical and operational constraints determine whether reported adversarial robustness can move beyond laboratory validity. Robustness claims become stronger when they are tested across modern and diverse datasets, evaluated through realistic NIDS pipelines, accompanied by system-level profiling, and validated over time under changing traffic and adaptive deployment conditions. Without these checks, reported improvements may remain confined to controlled benchmarks rather than demonstrating deployable, scalable, and durable security.

X. LESSONS LEARNED

This section distills the main conclusions that emerge from the reviewed literature on adversarial generation, evaluation, and defense in ML-based NIDS. Unlike the preceding discussion, which analyzes why these limitations occur, this section states the lessons that current evidence supports. The lessons are therefore backward-looking: they summarize what the field has learned so far, while the following roadmap translates these conclusions into future research directions.

Lesson 1: High adversarial success is an upper-bound signal, not proof of operational exploitability. Many attacks achieve high attack success rates in controlled feature-space settings, including white-box, black-box, and transfer-based scenarios. However, such success often reflects detector sensitivity under simplified assumptions rather than evidence of a deployable network threat. When protocol validity, functionality preservation, and replayability are enforced, the feasible perturbation space narrows and attack

effectiveness often decreases. The lesson is that high feature-space ASR should be interpreted as a diagnostic upper bound unless the study demonstrates that the adversarial traffic can be instantiated as valid, executable network behavior [90].

Lesson 2: Robustness is conditional on the evaluation setting, not intrinsic to a model family. The reviewed literature shows that traditional models, deep architectures, ensembles, graph-based detectors, and hybrid systems can all be vulnerable under adversarial manipulation. At the same time, no architecture family is uniformly robust or uniformly weak. Reported robustness depends on the dataset, preprocessing pipeline, feature representation, threat model, perturbation constraints, and metric used in the experiment. The lesson is that robustness in ML-based NIDS is not a fixed property of a classifier; it is a property of the full experimental and deployment context [5].

Lesson 3: Transferability is the sharpest test of pipeline dependence. Transferability is central to black-box evasion, but it is also the clearest case where evaluation design can shape the result. Many studies test transfer across models that share the same dataset, feature extraction process, and preprocessing routine. Under these conditions, high transfer success may reflect shared feature dependencies or pipeline artifacts rather than true architecture-independent vulnerability. The lesson is that transferability claims are credible only when evaluated across heterogeneous datasets, feature schemas, preprocessing pipelines, and model families [63].

Lesson 4: Perturbation constraints determine what kind of claim a study can support. The constraint regime is not a minor implementation detail; it determines the meaning of the robustness evidence. Unconstrained and norm-bounded attacks mainly test sensitivity to synthetic feature distortions. Protocol-aware, functionality-preserving, and packet-level executable attacks provide stronger evidence of operational adversarial risk. Similarly, defenses evaluated only against weakly constrained feature-space perturbations may appear more effective than they would under realistic traffic manipulations. The lesson is that every robustness claim should be interpreted through the admissibility level of the perturbations used to produce it.

Lesson 5: Current defenses demonstrate component robustness more often than deployable resilience. Adversarial training, detection and filtering, purification, robust architectures, ensembles, and hybrid frameworks can improve robustness under selected conditions. Yet their effectiveness usually depends on alignment among the attack type, feature representation, model architecture, dataset, and evaluation setup. Layered defenses can increase attacker effort, but layering alone does not guarantee system-level protection and may introduce coordination failures, shared preprocessing weaknesses, calibration burden, or latency. The lesson is that improved adversarial accuracy under a fixed benchmark should not be treated as evidence of deployable resilience [142], [173], [192].

TABLE 10. List of abbreviations and named frameworks used in this review.

Abbrev.	Meaning	Abbrev.	Meaning	Abbrev.	Meaning
General domain, explainability, feature processing, and data-balancing terms					
CAN	Controller Area Network	ICA	Independent Component Analysis	IG	Integrated Gradients
LIME	Local Interpretable Model-Agnostic Explanations	PCA	Principal Component Analysis	RFE	Recursive Feature Elimination
SHAP	Shapley Additive Explanations	SMOTE	Synthetic Minority Over-sampling Technique	XAI	Explainable Artificial Intelligence
Classical machine-learning and ensemble models					
AB	AdaBoost	Bagging	Bootstrap Aggregating	BNB	Bernoulli Naive Bayes
C4.5	C4.5 Decision Tree	CatBoost	Categorical Boosting	DT	Decision Tree
EBM	Explainable Boosting Machine	GB	Gradient Boosting	GBDT	Gradient-Boosted Decision Tree
GMM	Gaussian Mixture Model	GNB	Gaussian Naive Bayes	HGB	Histogram-based Gradient Boosting
HT	Hoeffding Tree	ID3	ID3 Decision Tree Algorithm	IF	Isolation Forest
KNN	K-Nearest Neighbors	KNNADWIN	k-NN with Adaptive Windowing	LDA	Linear Discriminant Analysis
LGBM	Light Gradient Boosting Machine	LOF	Local Outlier Factor	LR	Logistic Regression
LSVM	Linear Support Vector Machine	NB	Naive Bayes	OCSVM	One-Class Support Vector Machine
OneR	One-Rule Classifier	QDA	Quadratic Discriminant Analysis	RBF-SVM	Radial Basis Function Support Vector Machine
RF	Random Forest	SGD	Stochastic Gradient Descent	SMR	Soft-Max Regression
SOM	Self-Organizing Map	SVC	Support Vector Classifier	SVM	Support Vector Machine
XGB	eXtreme Gradient Boosting				
Deep, neural, graph, transformer, and generative architectures					
AE	Autoencoder	BiGAN	Bidirectional Generative Adversarial Network	BiGRU	Bidirectional Gated Recurrent Unit
BiLSTM	Bidirectional Long Short-Term Memory	BNN	Bayesian Neural Network	CapsNet	Capsule Network
CNN	Convolutional Neural Network	CTGAN	Conditional Tabular GAN	DAE	Denoising Autoencoder
DAGMM	Deep Autoencoding Gaussian Mixture Model	DBN	Deep Belief Network	DNN	Deep Neural Network
DSVDD	Deep Support Vector Data Description	E-GraphSAGE	Edge Graph Sample and Aggregation	E-ResGAT	Edge Residual Graph Attention Network
FCNN	Fully Connected Neural Network	FNN	Feedforward Neural Network	GAN	Generative Adversarial Network
GCN	Graph Convolutional Network	GNN	Graph Neural Network	GRU	Gated Recurrent Unit
JK-Net	Jumping Knowledge Network	LAE	Linear Autoencoder	LineGraphSAGE	Line Graph Sample and Aggregation
LSTM	Long Short-Term Memory	LuNET	LuNET Deep Learning NIDS Architecture	MLP	Multi-Layer Perceptron
NN	Neural Network	RBM	Restricted Boltzmann Machine	RCNN	Recurrent Convolutional Neural Network
RNN	Recurrent Neural Network	SAE	Stacked Autoencoder	SNN	Self-Normalizing Neural Network
TabNet	Attention-Based Tabular Deep Learning Model	TF	Transformer	Traffic-BERT	Traffic Bidirectional Encoder Representations from Transformers
TS-IDS	Temporal Sequence IDS	VAE	Variational Autoencoder	Var-CNN	Variant CNN
WGAN	Wasserstein GAN	WnD	Wide and Deep Network		
Reinforcement-learning, evolutionary, heuristic, and search methods					
A2C	Advantage Actor-Critic	DDQN	Double Deep Q-Network	DQN	Deep Q-Network
DRQN	Deep Recurrent Q-Network	GA	Genetic Algorithm	HBO	Harris Hawks Optimization
HW-NAS	Hardware-Aware Neural Architecture Search	NES	Natural Evolution Strategies	PPO	Proximal Policy Optimization
PSO	Particle Swarm Optimization	QRDQN	Quantile Regression Deep Q-Network	RL	Reinforcement Learning
SARSA	State–Action–Reward–State–Action	SE-DQN	Strength-Enhanced Deep Q-Network		
Adversarial attack and perturbation-generation methods					
A2PM	Adaptive Perturbation Pattern Method	ACG	Adversarial Constraint Generation	AFP	Adaptive Feature Poisoning
ANI	Adversarial Noise Injection	AOI	Adversarial Outlier Injection	AutoAttack	AutoAttack Ensemble Attack
BFAM	Brute-Force Attack Method	BIM	Basic Iterative Method	CW	Carlini & Wagner Attack
DF	DeepFool	DTA	Decision Tree Attack	EN	Elastic-Net Attack
FGM	Fast Gradient Method	FGSM	Fast Gradient Sign Method	HSJ	HopSkipJump Attack
IFGSM	Iterative Fast Gradient Sign Method	JSMA	Jacobian-Based Saliency Map Attack	L-BFGS	Limited-Memory BFGS Attack
MIM	Momentum Iterative Method	NF	NewtonFool	OPT-Attack	Optimization-Based Attack
PGD	Projected Gradient Descent	QGAN	Quantum Generative Adversarial Network	SPSA	Simultaneous Perturbation
TDTCE	Traffic Diffusion Models with Transformers against NIDS in Constrained Environments	UAP	Universal Adversarial Perturbation	ZOO	Stochastic Approximation Attack Zeroth-Order Optimization Attack
Defense, detection, robustness, and adversarial-training mechanisms					
AAD	Adversarial Attack Detector	AT	Adversarial Training	IBP	Interval-Bound Propagation
LARAR	Layer-wise Adversarial Robustness using Adaptive Regularization	QSVM	Quantum Support Vector Machine	RRCF	Robust Random Cut Forest

Lesson 6: Robustness claims are only as trustworthy as their assumptions are auditable. Across attack, evaluation, and defense studies, many results are difficult to reproduce

or compare because key assumptions and implementation details are underreported. These include attacker knowledge, feedback type, query budget, preprocessing steps, feature

TABLE 11. Reviewed studies by domain.

Ref	Dataset	Attacks / Perturbations	Evaluated Models	Defense / Detection
Generation — Black-Box & Transferable Attacks				
[26]	Kitsune	Substitute Model + Saliency-guided Iterative FGSM	Kitsune	None
[31]	CICIDS2017	Substitute Model + Iterative Perturb.	DT, RF, LSTM, AB, Bagging, Ens.	AT
[28]	CICIDS2017, Kitsune	Substitute Model + PGD Protocol-Aware Morphing	LR, SVM, NB, DT, RF, XGB, FNN, Kitsune	AT
[32]	CICIDS2018	Manifold Approximation + Spherical Local Subspace	DAGMM, AE, AnoGAN, ALAD, DSVDD, OCSVM, IF	None
[27]	Tor-nonTor	Substitute Model + Mutual Information-guided Query	DNN, SVM	None
[29]	CICIDS2017, MTA	FAT	LR, DT, SVM, MLP, IF, LSTM, FS-Net	None
[30]	UNSW-SOSR2019	HAA	GCN, JK-Net	None
[33]	OPC UA, MedBIoT	KNNBA; RandomBA; BFAM	RNN, MLP, DT, LR, RF	None
[35]	NSL-KDD, CICIDS2017	R-FGSM; R-BIM; R-PGD; R-MIM; U-ASG (+ LAE-based variants)	DAE, DAGMM, VAE, SAE, LAKE	None
[36]	NSL-KDD, Gas Pipeline, Edge-IIoTset	UAPG-IIoT; EUAPG-IIoT (+ FGSM, PGD, MIM, DF, CW bases)	NN, SVM, LR, DT, NB, RF	None
[34]	CICIDS2018, NSL-KDD	Silent Probing; Change-Point Detection; Causality-guided Perturbation	RF, HGB, HGB+MLP	IF (detectability only)
Generation — Domain-Constrained and Realistic Attacks				
[37]	CICIDS2017, IoT-23	A2PM	MLP, RF	AT
[39]	CTU-13, CICIDS2018	Adv-Bot	MLP, RF, KNN	Grouped-Feature Ens. Detector
[45]	ISCVPN2016	ANT	CNN, SAE	None
[40]	NSL-KDD, Gas Pipeline	FGSM; PGD; MIM; CW; DF (+ SAM, EAM bases)	NN, LR, DT, SVM, NB, RF	None
[42]	CTU-13	Flow-Feature Increment	RF	None
[46]	IoT Testbed, Kitsune	Liuer Mihou (PSO-DE)	Kitsune, SOM, RRCE, LOF, OCSVM	Feature Squeezing; MagNet
[50]	CTU-13, ToN-IoT	Structure-Preserving GNN	inductive GNN	None
[38]	CICIDS2017, NSL-KDD	Constrained Weighted CW	DNN, DBN, AE	None
[43]	IoT-23, Bot-IoT	A2PM	RF, XGB, LGBM, IF	Constraint-Aware AT
[47]	CICIDS2017	Packet Split; Delay; Injection	Kitsune, DAGMM, BiGAN	None
[48]	Testbed	Traffic-Control Shaping	BNN, DT, DNN, OneR	None
[49]	Kitsune, CICIDS2017	TANTRA	AE, KitNET, IF	Timing-Aware Detector
[41]	CICIDS2018	UAP; Customized UAP; FGSM; BIM	DQN-IDS	None
[51]	ToN-IoT, CICIDS2018-v2	GIANT	E-GraphSAGE, TS-IDS, Anomal-E, E-ResGAT, LineGraphSAGE	None
[52]	CIC Modbus 2023	Jitter; Padding	DT, RF, HGB	None
[44]	CICIDS2017, CICIDS2018, real-world smart-home network; CIC-BCCC-NRC/ACI-IoT-2023	Host-space perturbations	DT, RF, HGB, SVM, DNN, AE, RNN, LLM	AT
Generation — Evolutionary and Heuristic Algorithms				
[53]	CICIDS2017	GA; PWPSA	LSTM	None
[54]	NSL-KDD, UNSW-NB15	GA; PSO; GAN; Monte-Carlo	SVM, DT, NB, KNN, RF, MLP, GB, LR, LDA, QDA, Bagging, Ens.	None
[56]	CICIDS2017	Constrained GA Payload Substitution	CNN, FNN, AB	None
[58]	Kitsune, CICIDS2017	GAN+PSO; Random-Dup; Random-ST; TWA	KitNET, LR, DT, SVM, MLP, IF	AT; Feature Selection; Adversarial Feature Reduction
[57]	NSL-KDD	GA	DT	None
[55]	CICIDS2017	GA; PSO + live-network replay	RF, DT, SVM	None
Generation — Explainability-Guided Attacks				
[63]	CICIDS2017, Kitsune	ETA; JSM; CW; ZOO; HSJ; GAN	MLP, AlertNet, IDSNet, DeepNet, RF, XGB, MaMPF, FS-Net, KitNET, Diff-RF	None
[59]	CICIDS2017	XAI-driven Integrated Gradients	DNN	None
[60]	CICIDS2017, ToN-IoT	KernelSHAP-guided XAI	DNN	None
[61]	X-IIoTID	DeepSHAP-guided masked-PGD; FGSM; PGD; DeepFool; JSMA	DNN-NIDS	VAE reconstruction error
[62]	InSDN	SHAP-guided perturbations	2D-CNN	None
Generation — Generative Models				
[64]	CICIDS2017, Kitsune	MACGAN	IF, GMM, SVM, SAE, RBM, Kitsune	None
[70]	NSL-KDD	attackGAN; FGSM; PGD; CW	SVM, DT, RF, NB, DNN	None
[80]	Kitsune, CICIDS2017, MAWI-Lab	FPAttack	KitNET, LR, MLP, IF	FPDefense
[74]	NSL-KDD	GAN	KNN, DT, RF, SVM, NN	AT
[68]	NSL-KDD, CICIDS2018	DIGFuPAS	SVM, NB, MLP, LR, DT, RF, KNN, CNN, RNN	DIGFuPAS-based AT
[65]	CICIDS2017, CICIDS2018	DEMGAN	DT, NB, LR, MLP, CNN	AT
[79]	NSL-KDD, UNSW-NB15, CICDDoS2019	NIDFSM	SVM, LR, KNN, DT, RF, 3L-IDS, LUNET, CNN-BiLSTM	None
[77]	CICIDS2017	Feature-guided VAE Local-Search	RF, LR, NB, DT, KNN, SVM, GB, DNN	None
[71]	CICIDS2017	NIDSGAN	DNN, DT, SVM, KNN, LR	None
[76]	CICIDS2017	Gen-AAL	GBDT	None
[69]	NSL-KDD	IDS-GAN	DeepNet, MLP	None
[73]	KDDCup99	GAN	DNN, LR, SVM, KNN, NB, RF, DT, GB	GAN-based AT
[67]	NSL-KDD	JSMA; FGSM; DF; CW; Unrestricted GAN; Static GAN; IDSGAN	SVM, NB, MLP, LR, DT, RF, KNN	None

TABLE 11. (Continued). Reviewed studies by domain.

Ref	Dataset	Attacks / Perturbations	Evaluated Models	Defense / Detection
[66]	NSL-KDD, CICIDS2017	MasqueradeGAN-GP	DT, KNN, LR, MLP, NB, RF, SVM, Kitsune, DNN, Fed-anids	None
[81]	CICIDS2018, CICIDS2017, UNSW-NB15	ProGen	KNN, RF, XGB, CNN, LSTM, TF	None
[75]	CICIDS2017	DIGFuPAS	LR, SVM, NB, DT, RF, KNN	AT
[78]	NSL-KDD, CICIDS2017	TEAM; PGD; CW; IDSGAN; J-Attack	RNN, LSTM, GRU	None
[72]	UNSW-NB15	QGAN; vs. GAN, VAE, SMOTE	RF, MLP	None
[82]	CTU-Malware, CICIDS2012/2017, CIRA-CIC-DoHBRW-2020	TDTCE; PGA; PBA	LR, SVM, NB, DT, RF, KNN, MLP, CNN, Var-CNN, BiLSTM, BiGRU, LuNET	AT
[83]	UNSW-NB15, CICIDS2017, CICDDoS2019	NetDiffuser; FGSM; PGD; ACG	MLP-1L, MLP-5L, CNN-2L, CNN-3L	MANDA; Artifact for detectability testing
Generation — Gradient-Based Methods				
[88]	NSL-KDD	C-IFGSM; IFGSM	DNN, CNN, MLP	None
[89]	Electra	SIGSM; FGSM; BIM	DNN	None
Generation — Reinforcement-Learning Approaches				
[84]	CTU-13, ISOT	RL-based Flow Mutation	BotCatcher (CNN+LSTM), XGB	None
[85]	CICIDS2018, Live Traffic	SE-DQN Packet Mutation	AB, MLP, DT, LR, RF, SVM	Adaptive Decision-Threshold Adjustment
[87]	WIDE MAWI 2023	NetMasquerade	Whisper, FlowLens, NetBeacon, Vanilla+RNN, CICFlowMeter+MLP, Kitsune	None
Evaluation — Domain & Architecture-Specific Analyses				
[125]	NSL-KDD	FGSM, BIM	Deep-RL	None
[118]	NSL-KDD, CICIDS2017	FGSM, DF, CW, JSMA	DT, RF, NB, SVM, NN, DAE	AT
[117]	ISCX VPN-NonVPN, NIMS	ZOO, PGD, DF	C4.5, KNN, NN, CNN, RNN	None
[123]	SDN Traffic	FGSM, JSMA, JSMA-RE	MLP, CNN, LSTM	None
[129]	UNSW-NB15, Bot-IoT	JSMA, FGSM, CW	CNN, LSTM, DBN	None
[119]	IoTID20	JSMA, FGSM, DF	RF, DT, KNN, MLP	AT
[124]	NSL-KDD, UNSW-NB15, CICIoV2024	FGSM, BIM	DQN, PPO, TRPO, A2C, QRQDN	None
[127]	Bot-IoT	FGSM, BIM, PGD	FNN, SNN	None
[128]	IEEE-BigData19	FGSM	NN	None
[121]	CICIDS2018; MNIST	FGSM (varying ϵ)	DNN	None
[122]	USTC-TFC2016	FGSM, PGD	HW-NAS DNN	Adversarial fine-tuning
[126]	NSL-KDD	FGSM, BIM	DQN, DRQN, DNN	None
[130]	CICIDS2017	CW-L ₂	CapsNet NIDS	None
[131]	UNSW-NB15, CICIDS2017, CICIoT2023	FGSM, PGD, AutoAttack	1D-CNN + BiLSTM + Transformer	PGD-AT
[132]	CICIDS2017	HopSkipJump	XGBoost	LIME analysis
[120]	CICIDS2018	FGSM, PGD, BIM	DNN	None
Evaluation — Effectiveness & Benchmarking & Transferability				
[100]	CICIDS2017	VAM	AE, DT	None
[97]	NSL-KDD	FGSM, JSMA, DF	NB, LR, NN, MLP	None
[208]	UNSW-NB15, Bot-IoT, CICIDS2018	FGSM, JSMA, DF	SVM, RF, NB, CNN, RNN	None
[112]	NSL-KDD	FGSM, PGD, Boundary	MLP, SGD, SVM, RF	None
[111]	NSL-KDD	Substitute-Model CW, ZOO, GAN	DNN	None
[92]	NSL-KDD, UNSW-NB15	FGSM, BIM, PGD, JSMA, DF, CW, ZOO, Boundary, HSJ	AB, Bagging, GB, LR, DT, RF, SVC, DNN	Gaussian Augmentation
[95]	UNSW-NB15, Bot-IoT	JSMA, FGSM, CW	RF, SVM, DT, MLP	None
[109]	NewCICIDS, HIKARI	A2PM	RF, XGB, LGBM, EBM	AT
[115]	NSL-KDD	FGSM, PGD	DNN	None
[94]	NSL-KDD	PGD, MI-FGSM, L-BFGS, SPSA	DNN, SVM, RF, LR	None
[114]	CICIDS2017, CICDDoS2019	FGSM	DNN	None
[113]	UNSW-NB15	PGD	NN, RF	None
[107]	KDDCup99	FGSM, BIM, JSMA, CW	DT, SVM, NN	None
[116]	NSL-KDD	FGSM	DNN	None
[110]	CICIDS2018, InSDN	WGAN-GP, TTUR, AdvGAN	LR, DT, MLP, CNN, LSTM	AT
[93]	MQTTset, WUSTL-IIoT, X-IIoTID, Edge-IIoTset	FGSM, BIM, PGD, DF, JSMA, EN, ZOO	NB, DT, RF, XGB, DNN, CNN, RNN	None
[99]	KDDCup99	GA, Gaussian Noise	RF	None
[96]	NSL-KDD	FGSM, JSMA	DNN, CNN, LSTM, SMR, LR, NB, KNN, DT, AB, RF, Linear-SVM, RBF-SVM	None
[106]	NSL-KDD	CW, FGSM, BIM, JSMA	SVM, NB, MLP, LR, RF, DT, KNN	None
[98]	CICIDS2017, TRAbID2017	JSMA, FGSM	MLP	None
[108]	UNSW-NB15	FGSM, JSMA, LSTM State-Poisoning	LSTM	AT
[101]	CICIDS2017; CICIDS2018	FGSM, PGD; HSJA, ZOO	RF, LR, MLP, CNN1D	None
[102]	CICIDS2017	FGSM, PGD	RF, LSTM	None
[103]	NSL-KDD, CICIDS2017	FGSM, PGD	DNN	None
[104]	CICIDS2018	FGSM, BIM, PGD, CW	CatBoost	None
[105]	UNSW-NB15	semantic-preserving perturbations	LR, SVM, DT, RF, GB, MLP	None
Evaluation — Realism-Constrained Viability				
[134]	NSL-KDD	FGSM, PGD, BIM, CW, JSMA, DF, ZOO	MLP, CNN, CNN-LSTM, CNN-BiLSTM, SVM, DT, RF	None
[137]	UNSW-NB15, IoT-23	ZOO, DF, CW	DNN, DBN, DT, GB	AT

TABLE 11. (Continued). Reviewed studies by domain.

Ref	Dataset	Attacks / Perturbations	Evaluated Models	Defense / Detection
[136]	NSL-KDD, UNSW-NB15, CIDDs-001	FGSM, BIM, JSMA, DF, CW	MLP	None
[90]	Bot-IoT, WSN-DS	FGSM, BIM, PGD, DF, CW, NF, EN, ZOO, HSJ, Boundary	DNN	None
[133]	CICIDS2018	Iterative Feature-Space Perturb.	MLP, CNN, FNN, AB	None
[135]	Controlled testbed	FGSM (4 ϵ values; transfer)	FNN, KNN, DT	None
Defense — Adversarial Training-Based Defenses				
[206]	CICIDS2017	FGSM, JSMA, PGD, CW	DNN	AT
[209]	NSL-KDD	FGSM, JSMA, CW	DNN	AT
[207]	NSL-KDD, UNSW-NB15, CICIDS2017	FGSM, JSMA	DT, RF, SVM, KNN, LR, NB, MLP	AT
[142]	NSL-KDD	FGSM, BIM, PGD	DNN	AT
[143]	NSL-KDD	FGM, FGSM, PGD, JSMA	AB, DNN, DT, GB, KNN, LR, NB, RF, SVM	AIDTF
[148]	BATADAL	PGD	AE	AT, Sensitivity Training
[141]	NSL-KDD, UNSW-NB15	FGSM, PGD, BIM, DF, CW	NN, CNN, RNN	AT
[139]	ToN-IoT	FGSM, PGD	DNN	AT
[138]	NSL-KDD	PGD, FGSM	MLP	Min-Max AT
[144]	CICIDS2018, ECU-IoFT	FGSM, BIM, CW	DT, RF, LR, SVM, GAN	RIDS-IoFT
[140]	NSL-KDD, UNSW-NB15	FGSM, PGD	DNN, RNN	AT
[145]	CICIDS2017	GAN	AE; GAN	Sanitization + SMOTE + GAN-AT
[146]	UNSW-NB15	FGSM, PGD	DNN + auxiliary classifiers	LARAR: vulnerable-layer regularization + auxiliary supervision
[147]	CICIDS2017, CICIDS2018	FGSM, BIM, PGD, CW, DeepFool, JSMA; ZOO, Boundary, Pointwise, HSJ, OPT-Attack, GA	Two-layer CNN	AT
Defense — Detection & Filtering of Adversarial Samples				
[149]	CICIDS2018, KDDCup99, ROSIDS23	FGSM, PGD, BIM	DNN + Reconstruction-based	READ
[152]	CICIDS2017	FGSM, BIM, PGD, CW	NN, RF, AB, SVM, KNN	Activation-based Detector
[156]	NSL-KDD	Boundary, HSJ	Seq2Seq, TabNet	NIDS-DEFEND
[160]	NSL-KDD	FGSM, PGD	DNN, SVM, DT, LR, RF, LDA, Ens.	Detect & Reject
[153]	CICIDS2017, CICDDoS2019	CTGAN	CNN, RNN, MLP, AE	DDM-CNN
[157]	MedBIoT, IoTID20	Relation-Preserving Perturb.	LSTM, RNN, DT	FGMD
[154]	NSL-KDD, CICIDS2017	FGSM, BIM, CW	MLP, LR, KNN, BNB, DT, SVM	MANDA
[158]	NSL-KDD, UNSW-NB15, CICIDS2017	FGSM, BIM, PGD, DF, Boundary	ML, DNN	NIDS-CBAD
[150]	CICIDS2017, NSL-KDD, UNSW-NB15	FGSM, BIM, PGD, JSMA, DF	DNN, AE	AE
[155]	CICIDS2017	FGSM, BIM, CW	MLP, RF	AdvAT
[159]	CICIDS2018	FGSM, BIM, PGD, DeepFool	MLP, CNN, LSTM; RF (discriminator)	AAD (validator + discriminator + Ensemble)
[151]	NSL-KDD	FGSM, BIM, PGD	AE + RF; surrogate MLP	AE-RF with reconstruction-error gating
Defense — Explainability & Formal Verification				
[162]	NSL-KDD	FGSM, PGD, DF	MLP, DT, RF, LSVM,	AT + SHAP-based Analysis
[163]	NSL-KDD	PGD	SVM	XAI-based AE
[161]	UQ-IoT	ℓ_p -norm Perturb.	AE, DAE, Kitsune	NIDS-Vis
[165]	CICIDS2017, UNSW-NB15	Norm-bounded Perturb.	MLP	IBP Verification
[164]	CICIoT2023	FGSM, PGD, DeepFool	DNN	SHAP fingerprinting + AE detector
Defense — Hybrid & Multi-Stage Frameworks				
[196]	CICDDoS2019, UNSW-NB15	FGSM, JSMA, CW	CNN	PGD+PIOA + Spatial Smooth + ICA/RFE
[195]	NSL-KDD, UNSW-NB15	Protocol-Aware GA-based	Ens. (LR, SVM, DT, RF), LSTM, MLP	AT + SMOTE + Feature Engineering + Fine-Tuning
[202]	NSL-KDD, CICIDS2018	FGSM, BIM, DF, CW	KNN, LR, BNB, DT, SVM, LS, DNN	DLL-IDS
[204]	X-IIoTID, RT-IoT2022	Noise Injection, Feature Perturb.	CNN, RCNN, RBM, RCNN+RBM	Noise Filtering + Adaptive Feature Extraction + Dual-System Classification
[197]	CICIDS2018	FGSM, PGD, DF, Square	DNN	AT + SHAP-guided Feature Selection
[199]	CICIDS2017, CICIDS2018, CICDDoS2019	ZOO, HSJA, WGAN	MLP, NB, RF, DT, LR, Apollon	Apollon
[210]	CICDDoS2019	CW	DNN	Gaussian Augmentation, Feature Squeezing
[194]	CICIDS2018	FGSM, BIM, DF, JSMA	DNN	Def-IDS
[200]	UNSW-NB15, WUSTL-IIoT	BIM, FGSM, PGD, DF, AutoPGD, ZOO	DNN, XGB selector	DYNAMITE
[192]	CICIDS2017	FGSM, PGD, CW, JSMA, ZOO, DTA	DT, XGB, LR, SVM, DNN, Ens.	ELAT
[198]	CICIDS2018	GA	RF, DT, SVM	GAN Discriminator Filter + AT
[201]	CICIDS2017, CICIDS2018, CICDDoS2019	ZOO, FGSM	RF, SVM, LR, DNN, IDS-Anta	IDS-Anta
[205]	CICIDS2018	ZOO, GAN-based	RF	PCA-based Feature Selection + GAN-based AT

TABLE 11. (Continued). Reviewed studies by domain.

Ref	Dataset	Attacks / Perturbations	Evaluated Models	Defense / Detection
[203]	IoTID20, CICIOT2023	FGSM, BIM, MIM, PGD, CW, GAN	ARF-ADWIN, ARFDDM, HT, KNNADWIN, SRPDDM, SRPADWIN, PWPAAE	MSAAD, RADP, MABTSE
[193]	UNSW-NB15, ToN-IoT, Edge-IIoTset	FGSM, CW, ZOO, HSJ	LR, SVM, MLP, RF, Stacking, Voting	RobEns
Defense — Input Purification & Feature-Space Defenses				
[166]	SNMP-MIB, UNSW-NB15	FGSM, BIM, MIM, Madry	NN, DAE	DAE
[169]	NSL-KDD	FGSM, PGD, MI-FGSM	DNN, BiGAN	BiGAN Reconstruction Filter
Defense — Input Purification & Feature-Space Defenses (continued)				
[171]	NSL-KDD, UNSW-NB15	FGSM, BIM, JSMA, DF, CW	Diffusion, DNN	Diffusion-based Purification
[167]	CICIDS2017	Packet-Timing/Splitting Perturb.	Kitsune-AE, Kitsune-GMM, DAGMM, BiGAN, RePO, RePO+	RePO, RePO+
[170]	CICIDS2017	FGSM, JSMA, BIM, DF, CW, PGD, Auto-PGD	CNN	APE_GAN++
[168]	CICIDS2017, CICIDS2018, NSL-KDD	FGSM, PGD, CW	CNN	Denosing AE preprocessing
Defense — Multiple Defenses				
[24]	CICIDS2018	Boundary, HSJ, Pointwise, Opt-Attack, NES	MLP, CNN, LSTM	Model Voting, Ens. AT, Query Detector
[211]	CICIDS2017, NSL-KDD	FGSM, PGD, DF	CNN, RF, SVM	AT, Distillation, Feature Squeezing, AE Purification
[212]	NSL-KDD	PSO, GA, GAN, PGD	DT, RF, LR, LDA, QDA, MLP	AT, Mutable-Feature Removal, Soft-Voting Ens., Outlier Alarms
Defense — Robust Model Architectures & Ensembles				
[180]	NSL-KDD, UNSW-NB15, AWID	JSMA	DRL-DQN + Ens. (RF, AB, GNB, KNN, QDA)	DRL-based context-adaptive IDS + DAE
[190]	NSL-KDD	FGSM, IDSGAN	K-Means, DT, AE, MLP	Robust Semi-supervised
[184]	Generic Dataset	GA, PSO	SVM	Hybrid Classical-Quantum SVM (QSVM)
[191]	NSL-KDD	JSMA	MLP, RNN-ADV	RNN-ADV
[172]	CTU-13	Flow-Feature Increment	RF, MLP, DT, AB, WnD	AppCon
[189]	Mirai	FGSM	ARGAN-IDS	ARGAN-IDS
[177]	CICIDS2017	JSMA	MLP	Hierarchical Learning
[188]	CTU-13	Flow-Feature Increment	RF	RF Distillation
[174]	KDDCup99, DARPA-2009, CICIDS2017, Hydra-SDN	Flow/Payload/Rate Perturb.	RF, SVM, LR, NN, Ens.	Ens. + Feature Reduction
[179]	CICIDS2017, InSDN	WGAN, CTGAN, PGD	LGBM	RAIDS
[173]	CICDarknet2020	FGSM, BIM, DF, Boundary	RF, KNN, DT, LR (meta)	Stacking Ens. + AE Detector + AE Denoiser
[185]	N-BaIoT, Kitsune, WUSTL-IIoT	FGSM, PGD, Random Noise, Semantic Perturb.	FCNN	REDNN
[182]	CICIDS2017	Packet-Size, Inter-Arrival Time Perturb.	GNN, MLP, AB, RF, ID3	GNN-based NIDS
[175]	CICIDS2017, UNSW-NB15	FGSM, PGD	CNN; CNN-AE	AT + input sanitization + denoising AE
[176]	NSL-KDD, CICIDS2017	FGSM, PGD, DeepFool, GAN	CNN, LSTM, AE, Transformer	AT + sanitization + anomaly filter
[181]	CICIDS2018, NSL-KDD	Silent probing; transfer-based; decision-boundary	RF, HGB, HGB+MLP	AFP: traffic profiling + adaptive feature perturbation
[187]	NF-UQ-NIDS-v2	Realizability-aware constrained evasion	TabTransformer, XGBoost, LightGBM, verifiable tree ensembles	Adversarially trained TabTransformer + certified tree ensembles
[186]	CICIOT2023	FGSM, PGD	EdgeGuardNet; CNN, GRU	EdgeGuardNet: dilated CNN + Transformer-GRU + AT
[178]	NSL-KDD, CICIDS2017, KDDCup99	FGSM; PGD	HHO + CNN-Transformer-BiLSTM; hierarchical ensemble	HHO feature selection + Focal Loss + AT + hierarchical ensemble
[183]	CICDDoS2019, AdvCICDDoS2019	AP, AOI, ANI, AB; FGSM, DeepFool	GNN, DNN, Ensemble	GNN + AT + SHAP/LIME

scaling, random seeds, perturbation budgets, success definitions, and runtime or compute costs. The lesson is that adversarial robustness evidence must be auditable: readers should be able to determine what the attacker could observe, what the model received, how perturbations were generated, and under what conditions the reported result holds [18].

Lesson 7: Laboratory robustness does not automatically translate into operational resilience. Many studies remain bounded by legacy datasets, narrow attack coverage, static offline evaluation, and limited system-level profiling. Real deployments involve changing traffic distributions, encrypted or sampled traffic, middleboxes, flow reconstruction,

throughput constraints, concept drift, retraining cycles, and adaptive adversarial behavior. The lesson is that robustness must be validated beyond laboratory snapshots: evidence becomes stronger when models are tested across modern datasets, realistic NIDS pipelines, operational resource constraints, and lifecycle conditions that reflect how deployed systems evolve over time.

XI. FUTURE RESEARCH ROADMAP

The lessons identified above show that adversarial robustness in ML-based NIDS cannot be advanced through stronger attacks, larger models, or isolated defenses alone. Progress

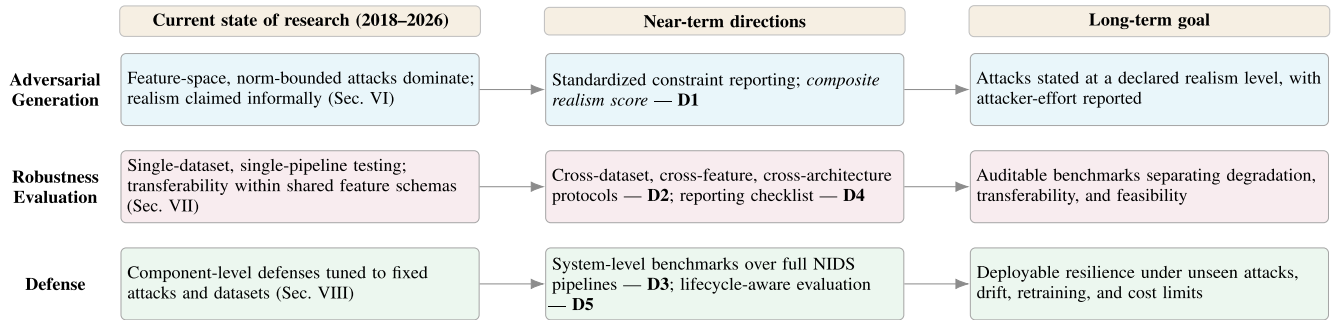


FIGURE 5. Research roadmap for adversarial evasion in ML-based NIDS, organized by the three review domains. Each lane maps the current state of the evidence to near-term directions (**D1**–**D5**, defined below) and to the long-term goal of operationally grounded resilience.

requires research programs that directly address the gap between benchmark performance and operational resilience. This roadmap translates the lessons learned into concrete future directions, linking each priority to the corresponding limitation identified in the evidence. Fig. 5 summarizes the roadmap across the three review domains, mapping the current state of research to near-term directions and long-term goals.

Direction (D1): Operationalize perturbation realism and constraint reporting. This direction responds to the lessons that high feature-space attack success is an upper-bound signal and that the constraint regime determines what kind of robustness claim a study can support. Future work should move from informal claims of realism toward explicit, standardized constraint reporting. Each study should state whether its perturbations are unconstrained, norm-bounded, feature-space constrained, protocol-aware, functionality-preserving, or packet-level executable. Beyond this classification, the field needs a *composite realism score* that combines protocol validity, malicious-functionality preservation, and replayability or emulation feasibility. Such a score would help distinguish diagnostic feature-space attacks from operationally executable attacks and would make adversarial results more comparable across generation, evaluation, and defense studies.

Direction (D2): Build heterogeneous evaluation protocols for robustness and transferability. This direction responds to the lessons that robustness is conditional on the evaluation setting and that transferability is credible only when pipelines genuinely differ. Future evaluation frameworks should require cross-dataset, cross-feature, cross-preprocessing, and cross-architecture testing. Transferability should not be reported only across models trained on the same dataset and feature representation; it should be evaluated across different feature extractors, preprocessing routines, traffic distributions, and model families. Future benchmarks should also report three evaluation axes separately: performance degradation, cross-pipeline transferability, and operational feasibility. This prevents a single ASR value from standing in for all three and reduces the risk that reported transferability reflects shared experimental artifacts rather than architecture-independent adversarial vulnerability.

Direction (D3): Develop system-level benchmarks for deployable defense resilience. This direction responds

to the lesson that current defenses often demonstrate component robustness rather than deployable resilience. Future defense research should evaluate complete NIDS pipelines rather than isolated modules. Benchmarks should test how adversarial training, filtering, purification, ensembles, and hybrid frameworks interact when placed into a shared detection workflow. They should also examine whether layered defenses introduce coordination failures, shared preprocessing weaknesses, calibration burden, or new attack surfaces. The goal should be to move from showing that a defense improves adversarial accuracy under a fixed attack to showing that the full pipeline remains robust across unseen attacks, feature representations, architectures, and operating conditions.

Direction (D4): Establish auditable reporting standards for adversarial NIDS research. This direction responds to the lesson that robustness claims are only as trustworthy as their assumptions are auditable. Future studies should adopt a common reporting checklist covering the threat-model class, attacker knowledge, feedback channel, query budget, perturbation budget, feature-to-traffic transformation path, preprocessing pipeline, random seeds, dataset splits, model configuration, success definition, runtime, and compute budget. Attack studies should report stealth and attacker-side cost using the indicators defined in Section IX-B, while defense studies should report system-side costs such as inference latency, throughput degradation, memory footprint, and energy consumption. This would make adversarial robustness claims easier to reproduce, compare, and audit.

Direction (D5): Move toward deployment-aware and lifecycle-aware robustness evaluation. This direction responds to the lesson that laboratory robustness does not automatically translate into operational resilience. Three aspects must be addressed together: the evaluation environment, the time horizon, and the resource profile. First, future work should evaluate adversarial robustness in environments that resemble deployed NIDS pipelines, including packet ingestion, flow reconstruction, traffic normalization, sampling, encrypted or partially visible traffic, middleboxes, proxies, load balancers, and real-time monitoring constraints. Second, evaluation should move beyond one-shot testing toward lifecycle-aware protocols that include concept drift, changing attack frequency, retraining cycles, threshold updates,

defense switching, and prolonged adversarial exposure. Third, computational efficiency should be treated as part of deployability: lightweight models, pruning, distillation, efficient architectures, and adaptive resource-aware defenses should be evaluated alongside robustness. A defense is not operationally robust unless it remains accurate, scalable, and maintainable over time.

Taken together, these directions define a shift from benchmark-centered adversarial robustness toward operationally grounded resilience. The next stage of adversarial NIDS research should not only ask whether an attack succeeds or a defense improves accuracy, but whether the resulting claim is realistic, transferable, auditable, deployable, and durable under changing network conditions.

XII. CONCLUSION

This review set out to separate adversarial evasion findings that reflect deployable, executable risk from those that reflect only diagnostic model sensitivity, and to translate that distinction into research directions. To this end, it synthesized 186 studies published between 2018 and 2026 on adversarial evasion in ML-based NIDS using a Kitchenham-informed systematic review protocol. Within a NIDS-exclusive and evasion-centered scope, the review introduced a lifecycle-aware, NIDS-specific threat model and organized the literature into three analytical domains: adversarial generation, robustness evaluation, and defense. Building on this structure, it developed a hierarchical taxonomy of methods, a two-tier limitation analysis, a perturbation-realism taxonomy ranging from feature-space manipulation to executable packet-level attacks, and an evidence-grounded set of lessons learned and future research directions.

The central conclusion is that technical progress in adversarial ML-based NIDS does not yet translate directly into operational robustness. Many reported evasion results remain diagnostic of model sensitivity under simplified feature-space assumptions, while many defenses demonstrate robustness only within narrow datasets, attacks, or evaluation settings. Advancing the field therefore requires treating robustness not as a static score measured on an isolated classifier, but as a lifecycle and system-level property shaped by realistic attacker capabilities, perturbation feasibility, heterogeneous pipelines, deployment cost, temporal drift, and adaptive defense requirements.

REFERENCES

- [1] Z. Ahmad, A. S. Khan, C. W. Shiang, J. Abdullah, and F. Ahmad, "Network intrusion detection system: A systematic study of machine learning and deep learning approaches," *Trans. Emerg. Telecommun. Technol.*, vol. 32, no. 1, p. 4150, Jan. 2021.
- [2] A. Aldweesh, A. Derhab, and A. Z. Emam, "Deep learning approaches for anomaly-based intrusion detection systems: A survey, taxonomy, and open issues," *Knowl.-Based Syst.*, vol. 189, Feb. 2020, Art. no. 105124.
- [3] A. Chakraborty, M. Alam, V. Dey, A. Chattopadhyay, and D. Mukhopadhyay, "A survey on adversarial attacks and defences," *CAAI Trans. Intell. Technol.*, vol. 6, no. 1, pp. 25–45, Mar. 2021.
- [4] N. Martins, J. M. Cruz, T. Cruz, and P. H. Abreu, "Adversarial machine learning applied to intrusion and malware scenarios: A systematic review," *IEEE Access*, vol. 8, pp. 35403–35419, 2020.
- [5] K. He, D. D. Kim, and M. R. Asghar, "Adversarial machine learning for network intrusion detection systems: A comprehensive survey," *IEEE Commun. Surveys Tuts.*, vol. 25, no. 1, pp. 538–566, 1st Quart., 2023.
- [6] M. elShehaby and A. Matrawy, "Evasion adversarial attacks remain impractical against ML-based network intrusion detection systems, especially dynamic ones," 2023, *arXiv:2306.05494*.
- [7] O. Ibitoye, R. Abou-Khamis, M. ElShehaby, A. Matrawy, and M. O. Shafiq, "The threat of adversarial attacks against machine learning in network security: A survey," *J. Electron. Electr. Eng.*, vol. 4, no. 1, pp. 16–59, Jan. 2025.
- [8] J. Lansky, S. Ali, M. Mohammadi, M. K. Majeed, S. H. T. Karim, S. Rashidi, M. Hosseinzadeh, and A. M. Rahmani, "Deep learning-based intrusion detection systems: A systematic review," *IEEE Access*, vol. 9, pp. 101574–101599, 2021.
- [9] I. J. Goodfellow, J. Shlens, and C. Szegedy, "Explaining and harnessing adversarial examples," 2014, *arXiv:1412.6572*.
- [10] A. Kurakin, I. Goodfellow, and S. Bengio, "Adversarial machine learning at scale," 2016, *arXiv:1611.01236*.
- [11] A. Madry, A. Makelov, L. Schmidt, D. Tsipras, and A. Vladu, "Towards deep learning models resistant to adversarial attacks," 2017, *arXiv:1706.06083*.
- [12] C. Szegedy, W. Zaremba, I. Sutskever, J. Bruna, D. Erhan, I. Goodfellow, and R. Fergus, "Intriguing properties of neural networks," 2013, *arXiv:1312.6199*.
- [13] S.-M. Moosavi-Dezfooli, A. Fawzi, and P. Frossard, "DeepFool: A simple and accurate method to fool deep neural networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 2574–2582.
- [14] N. Carlini and D. Wagner, "Towards evaluating the robustness of neural networks," in *Proc. IEEE Symp. Secur. Privacy (SP)*, May 2017, pp. 39–57.
- [15] N. Papernot, P. McDaniel, S. Jha, M. Fredrikson, Z. B. Celik, and A. Swami, "The limitations of deep learning in adversarial settings," in *Proc. IEEE Eur. Symp. Secur. Privacy (EuroS&P)*, Mar. 2016, pp. 372–387.
- [16] P.-Y. Chen, H. Zhang, Y. Sharma, J. Yi, and C.-J. Hsieh, "ZOO: Zeroth order optimization based black-box attacks to deep neural networks without training substitute models," in *Proc. 10th ACM Workshop Artif. Intell. Secur.*, Nov. 2017, pp. 15–26.
- [17] J. Chen, M. I. Jordan, and M. J. Wainwright, "HopSkipJumpAttack: A query-efficient decision-based attack," in *Proc. IEEE Symp. Secur. Privacy (SP)*, May 2020, pp. 1277–1294.
- [18] S. Ennaji, F. de Gaspary, D. Hitaj, A. Kbidi, and L. V. Mancini, "Adversarial challenges in network intrusion detection systems: Research insights and future prospects," *IEEE Access*, vol. 13, pp. 148613–148645, 2025.
- [19] I. Rosenberg, A. Shabtai, Y. Ellovici, and L. Rokach, "Adversarial machine learning attacks and defense methods in the cyber security domain," *ACM Comput. Surv.*, vol. 54, no. 5, pp. 1–36, Jun. 2022.
- [20] H. Khazane, M. Ridouani, F. Salahdine, and N. Kaabouch, "A holistic review of machine learning adversarial attacks in IoT networks," *Future Internet*, vol. 16, no. 1, p. 32, Jan. 2024.
- [21] A. Alotaibi and M. A. Rassam, "Adversarial machine learning attacks against intrusion detection systems: A survey on strategies and defense," *Future Internet*, vol. 15, no. 2, p. 62, Jan. 2023.
- [22] S. Saini, A. Chennamaneni, and B. Sawyerr, "A review of the duality of adversarial learning in network intrusion: Attacks and countermeasures," Dec. 2024, *arXiv:2412.13880*.
- [23] B. Kitchenham, *Procedures for Performing Systematic Reviews*, vol. 33. Keele, U.K.: Keele Univ., 2004, pp. 1–26.
- [24] C. Zhang, X. Costa-Perez, and P. Patras, "Adversarial attacks against deep learning-based network intrusion detection systems and defense mechanisms," *IEEE/ACM Trans. Netw.*, vol. 30, no. 3, pp. 1294–1311, Jun. 2022.
- [25] B. Biggio and F. Roli, "Wild patterns: Ten years after the rise of adversarial machine learning," in *Proc. ACM SIGSAC Conf. Comput. Commun. Secur.*, 2018, pp. 2154–2156.
- [26] H. Qiu, T. Dong, T. Zhang, J. Lu, G. Memmi, and M. Qiu, "Adversarial attacks against network intrusion detection in IoT systems," *IEEE Internet Things J.*, vol. 8, no. 13, pp. 10327–10335, Jul. 2021.
- [27] M. Usama, A. Qayyum, J. Qadir, and A. Al-Fuqaha, "Black-box adversarial machine learning attack on network traffic classification," in *Proc. 15th Int. Wireless Commun. Mobile Comput. Conf. (IWCMC)*, Jun. 2019, pp. 84–89.

- [28] H. Yan, X. Li, W. Zhang, R. Wang, H. Li, X. Zhao, F. Li, and X. Lin, "Automatic evasion of machine learning-based network intrusion detection systems," *IEEE Trans. Depend. Secure Comput.*, vol. 21, no. 1, pp. 153–167, Jan. 2024.
- [29] H. Sun, C. Peng, Y. Sang, S. Li, Y. Zhang, and Y. Zhu, "Evading encrypted traffic classifiers by transferable adversarial traffic," in *Collaborative Computing: Networking, Applications and Worksharing*, vol. 461, H. Gao, X. Wang, W. Wei, and T. Dagiuklas, Eds., Cham, Switzerland: Springer, 2022, pp. 153–173.
- [30] X. Zhou, W. Liang, W. Li, K. Yan, S. Shimizu, and K. I.-K. Wang, "Hierarchical adversarial attacks against graph-neural-network-based IoT network intrusion detection system," *IEEE Internet Things J.*, vol. 9, no. 12, pp. 9310–9319, Jun. 2022.
- [31] J. Sun and S. Yang, "Adversarial sample generation based on model simulation analysis in intrusion detection systems," *Electronics*, vol. 14, no. 5, p. 870, Feb. 2025.
- [32] A. Kuppa, S. Grzonkowski, M. R. Asghar, and N.-A. Le-Khac, "Black box attacks on deep anomaly detectors," in *Proc. 14th Int. Conf. Availability, Rel. Secur.*, Aug. 2019, pp. 1–10.
- [33] H. Jiang, J. Lin, and H. Kang, "KNNBA: A black-box adversarial attack against machine learning-based IoT NIDSs," *Int. J. Netw. Secur.*, vol. 27, no. 5, pp. 972–984, 2025.
- [34] S. Ennaji, E. Benkhelifa, and L. V. Mancini, "Vulnerability disclosure through adaptive black-box adversarial attacks in network intrusion detection systems," *Complex Intell. Syst.*, vol. 12, no. 1, p. 18, Jan. 2026.
- [35] Y. Zhang, Y. Wu, and X. Huang, "Toward transferable adversarial attacks against autoencoder-based network intrusion detectors," *IEEE Trans. Ind. Informat.*, vol. 20, no. 12, pp. 13863–13872, Dec. 2024.
- [36] S. Zhang, Y. Xu, and X. Xie, "Universal adversarial perturbations against machine-learning-based intrusion detection systems in industrial Internet of Things," *IEEE Internet Things J.*, vol. 12, no. 2, pp. 1867–1889, Jan. 2025.
- [37] J. Vitorino, N. Oliveira, and I. Praça, "Adaptive perturbation patterns: Realistic adversarial learning for robust intrusion detection," *Future Internet*, vol. 14, no. 4, p. 108, Mar. 2022.
- [38] M. Teuffenbach, E. Piatkowska, and P. Smith, "Subverting network intrusion detection: Crafting adversarial examples accounting for domain-specific constraints," in *Machine Learning and Knowledge Extraction*, vol. 12279, A. Holzinger, P. Kiesberg, A. M. Tjoa, and E. Weippl, Eds., Cham, Switzerland: Springer, 2020, pp. 301–320.
- [39] I. Debicha, B. Cochez, T. Kenaza, T. Debatty, J.-M. Dricot, and W. Mees, "Adv-bot: Realistic adversarial botnet attacks against network intrusion detection systems," *Comput. Secur.*, vol. 129, Jun. 2023, Art. no. 103176.
- [40] S. Zhang, Y. Xu, X. Zhang, and X. Xie, "Amplification methods to promote the attacks against machine learning-based intrusion detection systems," *Appl. Intell.*, vol. 54, no. 4, pp. 2941–2961, Feb. 2024.
- [41] H. Zhang, L. Zhang, G. Epiphaniou, and C. Maple, "A novel and practical universal adversarial perturbations against deep reinforcement learning based intrusion detection systems," 2025, *arXiv:2511.18223*.
- [42] G. Apruzzese and M. Colajanni, "Evading botnet detectors based on flows and random forest with adversarial samples," in *Proc. IEEE 17th Int. Symp. Netw. Comput. Appl. (NCA)*, Nov. 2018, pp. 1–8.
- [43] J. Vitorino, I. Praça, and E. Maia, "Towards adversarial realism and robust learning for IoT intrusion detection and classification," *Ann. Telecommun.*, vol. 78, nos. 7–8, pp. 401–412, Aug. 2023.
- [44] M. Verkerken, L. D'hooge, B. Volckaert, F. De Turck, and G. Apruzzese, "What is the problem space? Defining host-space adversarial perturbations against network intrusion detection systems," in *Proc. ACM Asia Conf. Comput. Commun. Secur. (ASIA CCS)*, 2026, pp. 1043–1059.
- [45] A. M. Sadehghzadeh, S. Shiravi, and R. Jalili, "Adversarial network traffic: Towards evaluating the robustness of deep-learning-based network traffic classification," *IEEE Trans. Netw. Service Manage.*, vol. 18, no. 2, pp. 1962–1976, Jun. 2021.
- [46] K. He, D. D. Kim, J. Sun, J. D. Yoo, Y. H. Lee, and H. K. Kim, "Liuer Mihou: A practical framework for generating and evaluating grey-box adversarial attacks against NIDS," Apr. 2022, *arXiv:2204.06113*.
- [47] M. J. Hashemi, G. Cusack, and E. Keller, "Towards evaluation of NIDSs in adversarial setting," in *Proc. 3rd ACM CoNEXT Workshop Big Data, Mach. Learn. Artif. Intell. Data Commun. Netw.*, Dec. 2019, pp. 14–21.
- [48] M. Catillo, A. Pecchia, A. Repola, and U. Villano, "Towards realistic problem-space adversarial attacks against machine learning in network intrusion detection," in *Proc. 19th Int. Conf. Availability, Rel. Secur.*, Jul. 2024, pp. 1–8.
- [49] Y. Sharon, D. Berend, Y. Liu, A. Shabtai, and Y. Elovici, "TANTRA: Timing-based adversarial network traffic reshaping attack," *IEEE Trans. Inf. Forensics Security*, vol. 17, pp. 3225–3237, 2022.
- [50] A. Venturi, D. Stabili, and M. Marchetti, "Problem space structural adversarial attacks for network intrusion detection systems based on graph neural networks," Apr. 2024, *arXiv:2403.11830*.
- [51] J. Zhao, D. Han, C. Ma, Q. Li, Z. Cui, H. Zhu, H. Zhang, M. He, Y. Lu, J. Dong, Y. Ma, and M. Shen, "GIANT: Structure-agnostic practical adversarial attacks for graph-based network intrusion detection systems," in *Proc. ACM Web Conf.*, 2026, pp. 3347–3357.
- [52] D. Galli, G. G. Zoccoli, D. Bianchini, D. Stabili, and M. Marchetti, "Evading ML network intrusion detection systems for Modbus TCP with problem-space perturbations," in *Proc. Joint Nat. Conf. Cybersecurity (ITASEC & SERICS)*, vol. 4198, 2026, pp. 1–17. [Online]. Available: <https://ceur-ws.org/Vol-4198/paper25.pdf>
- [53] W. Huang, X. Peng, Z. Shi, and Y. Ma, "Adversarial attack against LSTM-based DDoS intrusion detection system," in *Proc. IEEE 32nd Int. Conf. Tools with Artif. Intell. (ICTAI)*, Nov. 2020, pp. 686–693.
- [54] E. Alhajjar, P. Maxwell, and N. Bastian, "Adversarial machine learning in network intrusion detection systems," *Expert Syst. Appl.*, vol. 186, Dec. 2021, Art. no. 115782.
- [55] U. Guntupalli, M. Pujari, J. Halim, and W. Sun, "Investigating the influence of black-box adversarial machine learning attacks in live network environments," in *Proc. 12th Int. Conf. Inf. Syst. Secur. Privacy*, 2026, pp. 462–473.
- [56] M. Chale, B. Cox, J. Weir, and N. D. Bastian, "Constrained optimization based adversarial example generation for transfer attacks in network intrusion detection systems," *Optim. Lett.*, vol. 18, no. 9, pp. 2169–2188, Dec. 2024.
- [57] R. Mogg, S. Enoch, and D. S. Kim, "A framework for generating evasion attacks for machine learning based network intrusion detection systems," in *Information Security Applications (WISA 2021)* (Lecture Notes in Computer Science). Cham, Switzerland: Springer, 2021, pp. 51–63.
- [58] D. Han, Z. Wang, Y. Zhong, W. Chen, J. Yang, S. Lu, X. Shi, and X. Yin, "Evaluating and improving adversarial robustness of machine learning-based network intrusion detectors," *IEEE J. Sel. Areas Commun.*, vol. 39, no. 8, pp. 2632–2647, Aug. 2021.
- [59] S. Okada, H. Jmila, K. Akashi, T. Mitsunaga, Y. Sekiya, H. Takase, G. Blanc, and H. Nakamura, "XAI-driven adversarial attacks on network intrusion detectors," in *Proc. Eur. Interdiscipl. Cybersecurity Conf.*, Jun. 2024, pp. 65–73.
- [60] S. Okada, H. Jmila, K. Akashi, T. Mitsunaga, Y. Sekiya, H. Takase, G. Blanc, and H. Nakamura, "Xai-driven black-box adversarial attacks on network intrusion detectors," *Int. J. Inf. Secur.*, vol. 24, no. 3, p. 103, Jun. 2025.
- [61] B. Chang Chung and G.-B. Han, "A DeepSHAP-based adversarial attack on machine learning-based network intrusion detection," *IEEE Access*, vol. 14, pp. 2566–2575, 2026.
- [62] G. Mangiacapre, A. Nascita, F. Cerasuolo, A. Montieri, and A. Pescapè, "When explanations help attackers: XAI-guided adversarial attacks to network intrusion detection systems," in *Proc. IFIP Netw. Conf. (IFIP Netw.)*, May 2026, pp. 1–6.
- [63] H. Zhang, D. Han, S. Zhuang, Z. Wang, J. Sun, Y. Liu, J. Liu, and J. Dong, "Explainable and transferable adversarial attack for ML-based network intrusion detectors," *IEEE Trans. Depend. Secure Comput.*, vol. 22, no. 5, pp. 5090–5107, Sep. 2025.
- [64] Y. Zhong, Y. Zhu, Z. Wang, X. Yin, X. Shi, and K. Li, "An adversarial learning model for intrusion detection in real complex network environments," in *Wireless Algorithms, Systems, and Applications*, vol. 12384, D. Yu, F. Dressler, and J. Yu, Eds., Cham, Switzerland: Springer, 2020, pp. 794–806.
- [65] D. Xu, Y. Lv, M. Wang, B. Zheng, J. Zhao, and J. Yu, "DEMGAN: A machine learning-based intrusion detection system evasion scheme," *Comput. Mater. Continua*, vol. 84, no. 1, pp. 1731–1746, 2025.
- [66] B. Dong, H. Wang, and R. Luo, "Masqueradegan-gp: A generative adversarial network framework for evading black-box intrusion detection systems," *Internet Technol. Lett.*, vol. 8, no. 6, p. e640, Jan. 2025.
- [67] Z. Lin, Y. Shi, and Z. Xue, "IDSGAN: Generative adversarial networks for attack generation against intrusion detection," in *Advances in Knowledge Discovery and Data Mining (PAKDD 2022)* (Lecture Notes in Computer Science), vol. 13282. Cham, Switzerland: Springer, 2022, pp. 79–91.

- [68] P. T. Duy, L. K. Tien, N. H. Khoa, D. T. T. Hien, A. G.-T. Nguyen, and V.-H. Pham, "DIGFuPAS: Deceive IDS with GAN and function-preserving on adversarial samples in SDN-enabled networks," *Comput. Secur.*, vol. 109, Oct. 2021, Art. no. 102367.
- [69] D. Wang, X. Wang, and J. Fei, "IDS-GAN: Adversarial attack against intrusion detection based on generative adversarial networks," in *Proc. 5th Int. Conf. Comput. Vis., Image Deep Learn. (CVIDL)*, Apr. 2024, pp. 1130–1134.
- [70] S. Zhao, J. Li, J. Wang, Z. Zhang, L. Zhu, and Y. Zhang, "AttackGAN: Adversarial attack against black-box IDS using generative adversarial networks," *Proc. Comput. Sci.*, vol. 187, pp. 128–133, Jan. 2021.
- [71] B.-E. Zolbayar, R. Sheatsley, P. McDaniel, M. J. Weisman, S. Zhu, S. Zhu, and S. Krishnamurthy, "Generating practical adversarial network traffic flows using NIDSGAN," Mar. 2022, *arXiv:2203.06694*.
- [72] I. P. Pathak, N. Yilmazer, and D. Pankajkumar Patel, "Quantum GAN-based adversarial attack generation for network intrusion detection system," in *Proc. IEEE Int. Carnahan Conf. Secur. Technol. (ICCST)*, Oct. 2025, pp. 1–6.
- [73] M. Usama, M. Asim, S. Latif, J. Qadir, and Ala-Al-Fuqaha, "Generative adversarial networks for launching and thwarting adversarial attacks on network intrusion detection systems," in *Proc. 15th Int. Wireless Commun. Mobile Comput. Conf. (IWCMC)*, Jun. 2019, pp. 78–83.
- [74] A.-G. Mari, D. Zinca, and V. Dobrota, "Development of a machine-learning intrusion detection system and testing of its performance using a generative adversarial network," *Sensors*, vol. 23, no. 3, p. 1315, Jan. 2023.
- [75] C. P. X. Qui, D. H. Quang, P. T. Duy, D. T. T. Hien, and V.-H. Pham, "Strengthening IDS against evasion attacks with GAN-based adversarial samples in SDN-enabled network," in *Proc. RIVF Int. Conf. Comput. Commun. Technol. (RIVF)*, Aug. 2021, pp. 1–6.
- [76] D. Shu, N. O. Leslie, C. A. Kamhoua, and C. S. Tucker, "Generative adversarial attacks against intrusion detection systems using active learning," in *Proc. 2nd ACM Workshop Wireless Secur. Mach. Learn.*, Jul. 2020, pp. 1–6.
- [77] V. Kumar, K. Kumar, and M. Singh, "Generating practical adversarial examples against learning-based network intrusion detection systems," *Ann. Telecommun.*, vol. 80, nos. 3–4, pp. 209–226, Apr. 2025.
- [78] Z. Liu, D. Ye, L. Tang, Y. Zhang, J. Deng, and W. Kuang, "TEAM: Temporal adversarial examples attack model against network intrusion detection system applied to RNN," *IEEE Trans. Netw. Sci. Eng.*, vol. 12, no. 4, pp. 3400–3415, Jul. 2025.
- [79] R. Zhang, S. Luo, L. Pan, J. Hao, and J. Zhang, "Generating adversarial examples via enhancing latent spatial features of benign traffic and preserving malicious functions," *Neurocomputing*, vol. 490, pp. 413–430, Jun. 2022.
- [80] Y. Zhu, L. Cui, Z. Ding, L. Li, Y. Liu, and Z. Hao, "Black box attack and network intrusion detection using machine learning for malicious traffic," *Comput. Secur.*, vol. 123, Dec. 2022, Art. no. 102922.
- [81] M. Wang, N. Yang, N. J. Forcade-Perkins, and N. Weng, "Pro-Gen: Projection-based adversarial attack generation against network intrusion detection," *IEEE Trans. Inf. Forensics Security*, vol. 19, pp. 5476–5491, 2024.
- [82] Z. Sui, F. Kang, and H. Shu, "TDTCE: Generating adversarial instances against NIDS via diffusion model," *Comput. J.*, vol. 69, no. 6, pp. 994–1014, Jun. 2026.
- [83] P. Kumar, A. Saleh Md Tayeen, S. Misra, H. Cao, J. Liu, Q. Gong, and J. Harikumar, "NetDiffuser: Deceiving DNN-based network attack detection systems with diffusion-generated adversarial traffic," 2026, *arXiv:2603.08901*.
- [84] J. Wang, L. Qixu, W. Di, Y. Dong, and X. Cui, "Crafting adversarial example to bypass flow-&ML-based botnet detector via RL," in *Proc. 24th Int. Symp. Res. Attacks, Intrusions Defenses*, Oct. 2021, pp. 193–204.
- [85] S. Tan, X. Zhong, Z. Tian, and Q. Dong, "Sneaking through security: Mutating live network traffic to evade learning-based NIDS," *IEEE Trans. Netw. Service Manage.*, vol. 19, no. 3, pp. 2295–2308, Sep. 2022.
- [86] Y. Hwang, J. Yun, H. Jang, T.-T.-H. Le, and J. Son, "Adversarial attack on CAN IDS leveraging multi-DDQN agent," *IEEE Access*, vol. 14, pp. 13445–13462, 2026.
- [87] Z. Liu, Y. Zhao, Z. Liu, Q. Li, C. Fu, G. Zhou, and K. Xu, "A hard-label black-box evasion attack against ML-based malicious traffic detection systems," in *Proc. Netw. Distrib. Syst. Secur. Symp.*, 2026.
- [88] Y. Wang, Y. Wang, E. Tong, W. Niu, and J. Liu, "A c-ifgsm based adversarial approach for deep learning based intrusion detection," in *Verification and Evaluation of Computer and Communication Systems*, vol. 12519, B. Ben Heddia, Y.-F. Chen, G. Liu, and Z. Yu, Eds., Cham, Switzerland: Springer, 2020, pp. 207–221.
- [89] Á. L. Perales Gómez, L. F. Maimó, A. H. Celdrán, F. J. García Clemente, and F. Cleary, "Crafting adversarial samples for anomaly detectors in industrial control systems," *Proc. Comput. Sci.*, vol. 184, pp. 573–580, Jan. 2021.
- [90] H. A. Alatawi and C. Morisset, "Realism versus performance for adversarial examples against DL-based NIDS," in *Proc. 38th ACM/SIGAPP Symp. Appl. Comput.*, Mar. 2023, pp. 1549–1557.
- [91] F. Pierazzi, F. Pendlebury, J. Cortellazzi, and L. Cavallaro, "Intriguing properties of adversarial ML attacks in the problem space," in *Proc. IEEE Symp. Secur. Privacy (SP)*, May 2020, pp. 1332–1349.
- [92] H. Jmila and M. I. Khedher, "Adversarial machine learning for network intrusion detection: A comparative study," *Comput. Netw.*, vol. 214, Sep. 2022, Art. no. 109073.
- [93] O. Gungor, E. Li, Z. Shang, Y. Guo, J. Chen, J. Davis, and T. Rosing, "Rigorous evaluation of machine learning-based intrusion detection against adversarial attacks," in *Proc. IEEE Int. Conf. Cyber Secur. Resilience (CSR)*, Sep. 2024, pp. 152–158.
- [94] Y. Peng, J. Su, X. Shi, and B. Zhao, "Evaluating deep learning based network intrusion detection system in adversarial environment," in *Proc. IEEE 9th Int. Conf. Electron. Inf. Emergency Commun. (ICEIEC)*, Jul. 2019, pp. 61–66.
- [95] Y. Pacheco and W. Sun, "Adversarial machine learning: A comparative study on contemporary intrusion detection datasets," in *Proc. 7th Int. Conf. Inf. Syst. Secur. Privacy*, 2021, pp. 160–171.
- [96] S. Srinivasan, S. K. V. R. A. Soman, and S. Kp, "Towards evaluating the robustness of deep intrusion detection models in adversarial environment," in *Security in Computing and Communications (SSCC 2019)* (Communications in Computer and Information Science), vol. 1208, Cham, Switzerland: Springer, 2020, pp. 111–120.
- [97] M. Mulimani, R. Rachh, and S. Kavatagi, "A comparative approach: Machine learning and adversarial learning for intrusion detection," in *Proc. Int. Conf. Paradigms Comput., Commun. Data Sci.*, R. P. Yadav, S. J. Nanda, P. S. Rana, and M.-H. Lim, Eds., Singapore: Springer, 2023, pp. 477–488.
- [98] M. A. Ayub, W. A. Johnson, D. A. Talbert, and A. Siraj, "Model evasion attack on intrusion detection systems using adversarial machine learning," in *Proc. 54th Annu. Conf. Inf. Sci. Syst. (CISS)*, Mar. 2020, pp. 1–6.
- [99] M. Pujari, M. C. Medde, and S. Butakov, "Towards evaluating the influence of adversarial black-box algorithms on network intrusion detection datasets," *Tech. Rep.*, May 2025.
- [100] M. Catillo, A. Del Vecchio, A. Pecchia, and U. Villano, "A case study with CICIDS2017 on the robustness of machine learning against adversarial attacks in intrusion detection," in *Proc. 18th Int. Conf. Availability, Rel. Secur.*, Aug. 2023, pp. 1–8.
- [101] S. Aoudi and H. Al-Aqrabi, "Evaluating adversarial robustness of AI intrusion detection systems using automated traffic generation," *Res. Square Preprint*, Dec. 2025, doi: [10.21203/rs.3.rs-8316582/v1](https://doi.org/10.21203/rs.3.rs-8316582/v1).
- [102] B. Armand, "Robustness evaluation of machine learning-based intrusion detection systems under adversarial attacks," *TechRxiv Preprint*, 2025, doi: [10.36227/techrxiv.176703967.72285079/v1](https://doi.org/10.36227/techrxiv.176703967.72285079/v1).
- [103] T. S. C. S. Gaddam, "Adversarial vulnerability analysis of deep neural network-based intrusion detection systems using FGSM and PGD attacks," in *Proc. Int. Conf. Intell. Innov. Technol. Comput., Electr. Electron. (IITCEE)*, Jan. 2026, pp. 1–6.
- [104] D. Sivkov, V. Korzhuk, R. Safiullin, O. F. Jeelani, and A. Vorobeva, "Adversarial robustness of ML-based intrusion detection systems," in *Proc. 14th Int. Symp. Digit. Forensics Secur. (ISDFS)*, Mar. 2026, pp. 1–6.
- [105] A. Kumar, A. Rajak, S. Bhardwaj, S. Aggarwal, N. Sharma, and A. Yadav, "Evaluating the robustness of machine learning models against adversarial cyber attacks," in *Proc. 5th Int. Conf. Innov. Practices Technol. Manage. (ICIPTM)*, Feb. 2026, pp. 1–4.
- [106] R. Duan, W. Zhao, Z. J. Luo, N. Wang, Y. Liu, and Z. Lu, "Understanding the ineffectiveness of the transfer attack in intrusion detection system," in *Network Security Empowered by Artificial Intelligence*, vol. 107, Y. Chen, J. Wu, P. Yu, and X. Wang, Eds., Cham, Switzerland: Springer, 2024, pp. 99–119.

- [107] K. Tahayori, S. Saad, M. Mamun, and S. Samet, "Feasibility of adversarial attacks against machine learning models," in *Proc. 7th Int. Conf. Wireless, Intell. Distrib. Environ. Commun.*, vol. 237, I. Woungang and S. K. Dhurandher, Eds., Cham, Switzerland: Springer, 2025, pp. 19–39.
- [108] D. A. Kulikov and V. V. Platonov, "Adversarial attacks on intrusion detection systems using the LSTM classifier," *Autom. Control Comput. Sci.*, vol. 55, no. 8, pp. 1080–1086, Dec. 2021.
- [109] J. a. Vitorino, M. Silva, E. Maia, and I. Praça, "An adversarial robustness benchmark for enterprise network intrusion detection," in *Foundations and Practice of Security*, vol. 14551, M. Mosbah, F. Sèdes, N. Tawbi, T. Ahmed, N. Boulahia-Cuppens, and J. Garcia-Alfaro, Eds., Cham, Switzerland: Springer, 2024, pp. 3–17.
- [110] P. T. Duy, N. H. Khoa, D. T. T. Hien, H. D. Hoang, and V.-H. Pham, "Investigating on the robustness of flow-based intrusion detection system against adversarial samples using generative adversarial networks," *J. Inf. Secur. Appl.*, vol. 74, May 2023, Art. no. 103472.
- [111] K. Yang, J. Liu, C. Zhang, and Y. Fang, "Adversarial examples against the deep learning based network intrusion detection systems," in *Proc. MILCOM IEEE Mil. Commun. Conf. (MILCOM)*, Oct. 2018, pp. 559–564.
- [112] K. Talty, J. Stockdale, and N. D. Bastian, "A sensitivity analysis of poisoning and evasion attacks in network intrusion detection system machine learning models," in *Proc. MILCOM IEEE Mil. Commun. Conf. (MILCOM)*, Nov. 2021, pp. 1011–1016.
- [113] V. Heydari and K. Nyarko, "Evaluating network intrusion detection models for enterprise security: Adversarial vulnerability and robustness analysis," in *Proc. 27th Int. Conf. Enterprise Inf. Syst.*, 2025, pp. 699–708.
- [114] H. Mohammadian, A. H. Lashkari, and A. A. Ghorbani, "Evaluating deep learning-based NIDS in adversarial settings," in *Proc. 8th Int. Conf. Inf. Syst. Secur. Privacy*, 2022, pp. 435–444.
- [115] X. Zhang, X. Zheng, and D. D. Wu, "Attacking DNN-based intrusion detection models," *IFAC-PapersOnLine*, vol. 53, no. 5, pp. 415–419, 2020.
- [116] A. Warzynski and G. Kolaczek, "Intrusion detection systems vulnerability on adversarial examples," in *Proc. Innov. Intell. Syst. Appl. (INISTA)*, Jul. 2018, pp. 1–4.
- [117] R. Maarouf, D. Sattar, and A. Matrawy, "Evaluating resilience of encrypted traffic classification against adversarial evasion attacks," in *Proc. IEEE Symp. Comput. Commun. (ISCC)*, Sep. 2021, pp. 1–6.
- [118] N. Martins, J. M. a. Cruz, T. Cruz, and P. H. Abreu, "Analyzing the footprint of classifiers in adversarial denial of service contexts," in *Progress in Artificial Intelligence*, vol. 11805, P. M. Oliveira, P. Novais, and L. P. Reis, Eds., Cham, Switzerland: Springer, 2019, pp. 256–267.
- [119] V. Pantelakis, P. Bountakas, A. Farao, and C. Xenakis, "Adversarial machine learning attacks on multiclass classification of IoT network traffic," in *Proc. 18th Int. Conf. Availability, Rel. Secur.*, Aug. 2023, pp. 1–8.
- [120] M. elShehaby and A. Matrawy, "A no-defense defense against gradient-based adversarial attacks on ML-NIDS: Is less more?" 2026, *arXiv:2605.18666*.
- [121] M. ElShehaby and A. Matrawy, "Exploring the effect of DNN depth on adversarial attacks in network intrusion detection systems," 2025, *arXiv:2510.19761*.
- [122] A. Chehade, E. Ragusa, P. Gastaldo, and R. Zunino, "Adversarial robustness of traffic classification under resource constraints: Input structure matters," in *Proc. Int. Symp. Netw., Comput. Commun. (ISNCC)*, Oct. 2025, pp. 1–6.
- [123] C.-H. Huang, T.-H. Lee, L.-H. Chang, J.-R. Lin, and G. Horng, "Adversarial attacks on SDN-based deep learning IDS system," in *Mobile and Wireless Technology 2018*, vol. 513, K. J. Kim and H. Kim, Eds., Singapore: Springer, 2019, pp. 181–191.
- [124] M. A. Merzouk, C. Neal, J. Delas, R. Yaich, N. Boulahia-Cuppens, and F. Cuppens, "Adversarial robustness of deep reinforcement learning-based intrusion detection," *Int. J. Inf. Secur.*, vol. 23, no. 6, pp. 3625–3651, Dec. 2024.
- [125] M. A. Merzouk, J. Delas, C. Neal, F. Cuppens, N. Boulahia-Cuppens, and R. Yaich, "Evading deep reinforcement learning-based network intrusion detection with adversarial attacks," in *Proc. 17th Int. Conf. Availability, Rel. Secur.*, Aug. 2022, pp. 1–6.
- [126] C. Rookard, "Evaluating adversarial resilience of deep reinforcement learning algorithms for network intrusion detection," in *Proc. IEEE 5th Int. Conf. AI Cybersecurity (ICAIC)*, Feb. 2026, pp. 1–6.
- [127] O. Ibtouye, O. Shafiq, and A. Matrawy, "Analyzing adversarial attacks against deep learning for intrusion detection in IoT networks," in *Proc. IEEE Global Commun. Conf. (GLOBECOM)*, Dec. 2019, pp. 1–6.
- [128] A. Piplai, S. S. L. Chukkappalli, and A. Joshi, "NAttack! adversarial attacks to bypass a GAN based classifier trained to detect network intrusion," in *Proc. IEEE 6th Int. Conf. Big Data Secur. Cloud (BigDataSecurity) Int. Conf. High Perform. Smart Comput., (HPSC) IEEE Int. Conf. Intell. Data Secur. (IDS)*, May 2020, pp. 49–54.
- [129] K. Albulayhi and Q. Abu Al-Haija, "Adversarial deep learning in anomaly based intrusion detection systems for IoT environments," *Int. J. Wireless Microw. Technol.*, vol. 13, no. 4, pp. 1–10, Aug. 2023.
- [130] M. Choudhary and K. Jain, "Evaluating a capsule network-based network intrusion detection system under Carlini–Wagner adversarial attacks," *Int. J. All Res. Educ. Sci. Methods*, vol. 14, no. 4, Apr. 2026, doi: 10.56025/IJARESM.2026.14042680.
- [131] J. A. González-Ramos, E. J. Abril, P. Fernández, J. Prieto, and P. Chamoso, "Adversarial robustness evaluation of hybrid CNN-LSTM-transformer NIDS on evolving threats," *J. Inf. Secur. Appl.*, vol. 100, Jul. 2026, Art. no. 104467.
- [132] H. Goncalves and Z. Alomari, "LIME-based analysis of adversarial feature manipulation in cloud intrusion detection systems," in *Proc. 2nd Int. Conf. Comput. Intell. Approaches Appl. (ICCIAA)*, May 2026, pp. 1–7.
- [133] S. Ennaji, E. Benkhefifa, and L. V. Mancini, "Toward realistic adversarial attacks in IDS: A novel feasibility metric for transferability," Apr. 2025, *arXiv:2504.08480*.
- [134] A. Grini, O. Taheri, B. El Khamlichi, and A. El Fallah-Seghrouchni, "Constrained network adversarial attacks: Validity, robustness, and transferability," in *Proc. 21st Int. Conf. Distrib. Comput. Smart Syst. Internet Things (DCOSS-IoT)*, Jun. 2025, pp. 771–776.
- [135] M. Catillo, A. Pecchia, and U. Villano, "Similarity is not enough: Issues with adversarial perturbations of traffic features against intrusion detection systems," in *Proc. 12th Int. Conf. Inf. Syst. Secur. Privacy*, 2026, pp. 325–332.
- [136] M. A. Merzouk, F. Cuppens, N. Boulahia-Cuppens, and R. Yaich, "Investigating the practicality of adversarial evasion attacks on network intrusion detection," *Ann. Telecommun.*, vol. 77, nos. 11–12, pp. 763–775, Dec. 2022.
- [137] N. Alhussien, A. Aleroud, A. Melhem, and S. Y. Khamaiseh, "Constraining adversarial attacks on network intrusion detection systems: Transferability and defense analysis," *IEEE Trans. Netw. Service Manage.*, vol. 21, no. 3, pp. 2751–2772, Jun. 2024.
- [138] S. Grierson, C. Thomson, P. Papadopoulos, and B. Buchanan, "Min-max training: Adversarially robust learning models for network intrusion detection systems," in *Proc. 14th Int. Conf. Secur. Inf. Netw. (SIN)*, vol. 1, Dec. 2021, pp. 1–8.
- [139] E. Lella, N. Macchiarulo, A. Pazeniza, D. Lofù, A. Abbatecola, and P. Novello, "Improving the robustness of DNNs-based network intrusion detection systems through adversarial training," in *Proc. 8th Int. Conf. Smart Sustain. Technol. (SpliTech)*, Jun. 2023, pp. 1–6.
- [140] K. Sato and P. Nair, "Improving intrusion detection robustness through adversarial training methods," *Frontiers Artif. Intell. Res.*, vol. 2, no. 3, pp. 457–482, Oct. 2025.
- [141] R. A. Khamis and A. Matrawy, "Evaluation of adversarial training on different types of neural networks in deep learning-based IDSs," in *Proc. Int. Symp. Netw., Comput. Commun. (ISNCC)*, Oct. 2020, pp. 1–6.
- [142] I. Debicha, T. Debatty, J.-M. Dricot, and W. Mees, "Adversarial training for deep learning-based intrusion detection systems," in *Proc. 16th Int. Conf. Syst. (ICONS)*, 2021, pp. 45–49.
- [143] W. D. Xiong, K. L. Luo, and R. Li, "AIDTF: Adversarial training framework for network intrusion detection," *Comput. Secur.*, vol. 128, May 2023, Art. no. 103141.
- [144] T. Gaber, T. Ali, M. Nicho, and M. Torky, "Robust attacks detection model for Internet of Flying Things based on generative adversarial network (GAN) and adversarial training," *IEEE Internet Things J.*, vol. 12, no. 13, pp. 23961–23974, Jul. 2025.
- [145] P. Chouragadey, Y. Sanodiya, and S. Ghosh, "Sanitization-driven, adversarial-informed framework for network anomaly detection," in *Proc. eNergetics*, Niš, Serbia: Complex System Research Centre, 2025.
- [146] H. Nasir, E. Javed, B. Shabir, Z. Jalil, and A. Mohsin, "Enhancing adversarial robustness in network intrusion detection: A layer-wise adaptive regularization approach," 2026, *arXiv:2605.08910*.
- [147] N. A. Roken, H. Hacid, A. Bouridane, and A. Hussian, "Hybrid adversarial retraining approach for a proactive network intrusion detection system," *IEEE Open J. Commun. Soc.*, vol. 7, pp. 5350–5364, 2026.

- [148] A. Flatscher, B. Stojanović, and O. Özdenizci, "Enhancing adversarial robustness of anomaly detection-based IDS in OT environments," in *Proc. 20th Int. Conf. Neww. Service Manage. (CNSM)*, Oct. 2024, pp. 1–7.
- [149] E. Değirmenci, I. Özçelik, and A. Yazıcı, "Adversarial attack detection approach for intrusion detection systems," *IEEE Access*, vol. 12, pp. 195996–196009, 2024.
- [150] V. Kumar, K. Kumar, M. Singh, and N. Kumar, "NIDS-DA: Detecting functionally preserved adversarial examples for network intrusion detection system using deep autoencoders," *Expert Syst. Appl.*, vol. 270, Apr. 2025, Art. no. 126513.
- [151] R. Ibrahim, A. A. Gani, N. Mohamed, M. Harbia, H. A. Mohammed, A. Kamal, and Z. Alhazza, "A hybrid autoencoder-random forest defense with reconstruction gating against adversarial evasion in intrusion detection systems," in *Proc. Int. Conf. Elect./Electron., Robot., Artif. Intell., Informat. (ICERAI)*, Apr. 2026, pp. 1–6.
- [152] M. Pawlicki, M. Choraś, and R. Kozik, "Defending network intrusion detection systems against adversarial evasion attacks," *Future Gener. Comput. Syst.*, vol. 110, pp. 148–154, Sep. 2020.
- [153] T.-T. Nguyen, T.-M. Bui, T.-T. Hoang, T.-V. Nguyen, X.-H. Nguyen, C.-S. Shieh, M.-F. Horng, and T.-L. Nguyen, "Detection of adversarial attacks using enhanced density metrics," *J. Inf. Hiding Multimedia Signal Process.*, vol. 16, no. 1, pp. 334–349, 2025.
- [154] N. Wang, Y. Chen, Y. Xiao, Y. Hu, W. Lou, and Y. T. Hou, "MANDA: On adversarial example detection for network intrusion detection system," *IEEE Trans. Depend. Secure Comput.*, vol. 20, no. 2, pp. 1139–1153, Mar. 2023.
- [155] T. Watanabe, E. Kodama, B. B. Bista, J. Wang, and T. Takata, "Proposal of adversarial attack traffic detection using adversarial attack techniques for network intrusion detection system," in *Advances in Network-Based Information Systems*, vol. 224, L. Barolli, Ed., Cham, Switzerland: Springer, 2024, pp. 174–186.
- [156] D. Khettaf and L. Bouzar-Benlabiod, "Defending the defender: Detecting adversarial examples for network intrusion detection systems," in *Proc. IEEE Int. Conf. Inf. Reuse Integr. Data Sci. (IRI)*, Aug. 2024, pp. 150–155.
- [157] H. Jiang, J. Lin, and H. Kang, "FGMD: A robust detector against adversarial attacks in the IoT network," *Future Gener. Comput. Syst.*, vol. 132, pp. 194–210, Jul. 2022.
- [158] V. Kumar, K. Kumar, and M. Singh, "NIDS-CBAD: Detecting adversarial attacks in network intrusion detection systems using domain constraints," *Int. J. Mach. Learn. Cybern.*, vol. 16, nos. 7–8, pp. 4213–4234, Aug. 2025.
- [159] P. Verma, A. Vidyarthi, A. Mishra, J. Ali, N. Bharot, and J. G. Breslin, "A secure adversarial attack detector framework for monitoring network intrusion in IoT environment," *IEEE Trans. Consum. Electron.*, vol. 71, no. 4, pp. 11613–11623, Nov. 2025.
- [160] I. Debicha, T. Debatty, J.-M. Dricot, W. Mees, and T. Kenaza, "Detect & reject for transferability of black-box adversarial attacks against network intrusion detection systems," in *Advances in Cyber Security (ACeS 2021)* (Communications in Computer and Information Science), vol. 1487, Cham, Switzerland: Springer, 2021, pp. 329–339.
- [161] K. He, D. D. Kim, and M. R. Asghar, "NIDS-vis: Improving the generalized adversarial robustness of network intrusion detection system," *Comput. Secur.*, vol. 145, Oct. 2024, Art. no. 104028.
- [162] K. Sauka, G.-Y. Shin, D.-W. Kim, and M.-M. Han, "Adversarial robust and explainable network intrusion detection systems based on deep learning," *Appl. Sci.*, vol. 12, no. 13, p. 6451, Jun. 2022.
- [163] E. Tcydenova, T. W. Kim, C. Lee, and J. H. Park, "Detection of adversarial attacks in ai-based intrusion detection systems using explainable AI," *Hum.-Centric Comput. Inf. Sci.*, vol. 11, p. 1, Sep. 2021.
- [164] D. P. Sharma, L. Xue, X. Sun, X. Lin, and P. Xiong, "Enhancing adversarial robustness of IoT intrusion detection via SHAP-based attribution fingerprinting," 2025, *arXiv:2511.06197*.
- [165] E. Nowroozi, R. Taheri, M. Hajizadeh, and T. Bauschert, "Verifying the robustness of machine learning based intrusion detection against adversarial perturbation," in *Proc. IEEE Int. Conf. Cyber Secur. Resilience (CSR)*, Sep. 2024, pp. 9–15.
- [166] Y. Aislman, M. Alkasasbeh, and M. Almseidin, "A robust SNMP-MIB intrusion detection system against adversarial attacks," *Arabian J. Sci. Eng.*, vol. 49, no. 3, pp. 4179–4195, Mar. 2024.
- [167] M. J. Hashemi and E. Keller, "Enhancing robustness against adversarial examples in network intrusion detection systems," in *Proc. IEEE Conf. Netw. Function Virtualization Softw. Defined Netw. (NFV-SDN)*, Nov. 2020, pp. 37–43.
- [168] G. D. Kamara, X. Wang, M. S. Alusine, and S. K. Dzokoto, "Robust CNN-based network intrusion detection against adversarial attacks using autoencoder input denoising," *Res. Square Preprint*, 2026, doi: 10.21203/rs.3.rs-9619720/v1.
- [169] Y. Peng, G. Fu, Y. Luo, J. Hu, B. Li, and Q. Yan, "Detecting adversarial examples for network intrusion detection system with GAN," in *Proc. IEEE 11th Int. Conf. Softw. Eng. Service Sci. (ICSESS)*, Oct. 2020, pp. 6–10.
- [170] A. Alotaibi and M. A. Rassam, "Enhancing the sustainability of deep-learning-based network intrusion detection classifiers against adversarial attacks," *Sustainability*, vol. 15, no. 12, p. 9801, Jun. 2023.
- [171] M. A. Merzouk, E. Beurier, R. Yaich, N. Boulahia-Cuppens, F. Cuppens, and F. Khomh, "Diffusion-based adversarial purification for intrusion detection," in *Data and Applications Security and Privacy XXXIX (DBSec2025)* (Lecture Notes in Computer Science). Cham, Switzerland: Springer, 2025, pp. 351–370.
- [172] G. Apruzzese, M. Andreolini, M. Marchetti, V. G. Colacino, and G. Russo, "AppCon: Mitigating evasion attacks to ML cyber detectors," *Symmetry*, vol. 12, no. 4, p. 653, Apr. 2020.
- [173] H. Mohanty, A. H. Roudsari, and A. H. Lashkari, "Robust stacking ensemble model for darknet traffic classification under adversarial settings," *Comput. Secur.*, vol. 120, Sep. 2022, Art. no. 102830.
- [174] A. Ganesan and K. Sarac, "Mitigating evasion attacks on machine learning based NIDS systems in SDN," in *Proc. IEEE 7th Int. Conf. Netw. Softwarization (NetSoft)*, Jun. 2021, pp. 268–272.
- [175] D. M. Musthafa, "Adversarial robustness in AI-driven cybersecurity solutions: Thwarting evasion assaults in real-time detection systems," *Int. J. Adv. Eng., Manage. Sci.*, vol. 11, no. 5, pp. 73–82, 2025.
- [176] M. H. A. Rai, Y. Noor, M. Faisal, and M. F. Nawazish, "Adversarial robustness of deep learning-based intrusion detection systems against AI-powered cyber attacks," *Spectr. Eng. Sci.*, vol. 3, no. 11, pp. 899–922, Nov. 2025.
- [177] A. McCarthy, E. Ghadafi, P. Andriotis, and P. Legg, "Defending against adversarial machine learning attacks using hierarchical learning: A case study on network traffic attack classification," *J. Inf. Secur. Appl.*, vol. 72, Feb. 2023, Art. no. 103398.
- [178] M. F. Alenezi and F. M. Alotaibi, "Adversarially robust hierarchical intrusion detection using hybrid deep learning and metaheuristic optimization," *Res. Square Preprint*, 2026, doi: 10.21203/rs.3.rs-9314868/v1.
- [179] A. Sankaya, B. G. Kılıç, and M. Demirci, "RAIDS: Robust autoencoder-based intrusion detection system model against adversarial attacks," *Comput. Secur.*, vol. 135, Dec. 2023, Art. no. 103483.
- [180] K. Sethi, E. Sai Rupesh, R. Kumar, P. Bera, and Y. Venu Madhav, "A context-aware robust intrusion detection system: A reinforcement learning-based approach," *Int. J. Inf. Secur.*, vol. 19, no. 6, pp. 657–678, Dec. 2020.
- [181] S. Ennaji, E. Benkhelifa, and L. V. Mancini, "Behavior-aware and generalizable defense against black-box adversarial attacks for ML-based IDS," 2025, *arXiv:2512.13501*.
- [182] D. Pujol-Perich, J. Suarez-Varela, A. Cabellos-Aparicio, and P. Barlet-Ros, "Unveiling the potential of graph neural networks for robust intrusion detection," *ACM SIGMETRICS Perform. Eval. Rev.*, vol. 49, no. 4, pp. 111–117, Jun. 2022.
- [183] U. G. Mohammad, A. Afzal, S. Alghamdi, A. Amjad, S. Yonbawi, M. Rizwan, O. Bagdasar, and K. Manuella, "An optimized graph neural network approach for robust and explainable IoT intrusion detection against adversarial attacks," *Sci. Rep.*, vol. 16, no. 1, p. 21453, May 2026.
- [184] D. Kumar Kejriwal, A. Goel, and A. Sharma, "Advancing adversarial robustness in cybersecurity: Gradient-free attacks and quantum-inspired defenses for machine learning models," *Int. J. Innov. Sci. Res. Technol. (IJISRT)*, vol. 10, no. 4, pp. 54–65, Apr. 2025.
- [185] I. Zakariyya, H. Kalutarage, and M. O. Al-Kadri, "Robust, effective and resource efficient deep neural network for intrusion detection in IoT networks," in *Proc. 8th ACM Cyber-Phys. Syst. Secur. Workshop*, May 2022, pp. 41–51.
- [186] Y. Shobha, K. Jeevarathinam, R. Saranya, V. R. Nomula, V. Velmurugan, and S. C. Madugula, "Adversarially-robust machine learning framework for real-time intrusion detection in edge-based IoT networks," in *Proc. 9th Int. Conf. Intell. Comput. Control Syst. (ICICCS)*, Mar. 2026, pp. 1–8.
- [187] H. A. Mukhef, "Adversarial robustness of network IDS (structured data)," *J. Al-Qadisiyah Comput. Sci. Math.*, vol. 17, no. 4, pp. 209–224, Dec. 2025.

- [188] G. Apruzzese, M. Andreolini, M. Colajanni, and M. Marchetti, "Hardening random forest cyber detectors against adversarial attacks," *IEEE Trans. Emerg. Topics Comput. Intell.*, vol. 4, no. 4, pp. 427–439, Aug. 2020.
- [189] J. Costa, F. Apolinário, and C. Ribeiro, "ARGAN-IDS: Adversarial resistant intrusion detection systems using generative adversarial networks," in *Proc. 19th Int. Conf. Availability, Rel. Secur.*, Jul. 2024, pp. 1–10.
- [190] S. M. M. Rashid, M. Toufikuzzaman, and M. S. Hossain, "A deep learning based semi-supervised network intrusion detection system robust to adversarial attacks," in *Proc. 10th Int. Conf. Netw., Syst. Secur.*, Dec. 2023, pp. 25–34.
- [191] A. U. H. Qureshi, H. Larijani, M. Yousefi, A. Adeel, and N. Mtetwa, "An adversarial approach for intrusion detection systems using Jacobian saliency map attacks (JSMA) algorithm," *Computers*, vol. 9, no. 3, p. 58, Jul. 2020.
- [192] Y.-D. Lin, J.-H. Pratama, D. Sudyana, Y.-C. Lai, R.-H. Hwang, P.-C. Lin, H.-Y. Lin, W.-B. Lee, and C.-K. Chiang, "ELAT: Ensemble learning with adversarial training in defending against evaded intrusions," *J. Inf. Secur. Appl.*, vol. 71, Dec. 2022, Art. no. 103348.
- [193] S. Alkadi, S. Al-Ahmadi, and M. M. Ben Ismail, "RobEns: Robust ensemble adversarial machine learning framework for securing IoT traffic," *Sensors*, vol. 24, no. 8, p. 2626, Apr. 2024.
- [194] J. Wang, J. Pan, I. AlQerm, and Y. Liu, "Def-IDS: An ensemble defense mechanism against adversarial attacks for deep learning-based network intrusion detection," in *Proc. Int. Conf. Comput. Commun. Netw. (ICCCN)*, Jul. 2021, pp. 1–9.
- [195] B. Tafreshian and S. Zhang, "A defensive framework against adversarial attacks on machine learning-based network intrusion detection systems," in *Proc. IEEE 23rd Int. Conf. Trust, Secur. Privacy Comput. Commun. (TrustCom)*, Dec. 2024, pp. 2436–2441.
- [196] K. Barik and S. Misra, "A comprehensive defense approach of deep learning-based NIDS against adversarial attacks," *Multimedia Tools Appl.*, vol. 84, no. 31, pp. 37745–37791, Jul. 2025.
- [197] H. Holla, S. R. Polepalli, and A. A. Sasikumar, "Adversarial threats to cloud IDS: Robust defense with adversarial training and feature selection," *IEEE Access*, vol. 13, pp. 84992–85003, 2025.
- [198] M. Pujari and W. Sun, "Fortifying machine learning-powered intrusion detection: A defense strategy against adversarial black-box attacks," in *Proc. 9th Int. Congr. Inf. Commun. Technol.*, vol. 1000, X.-S. Yang, S. Sherratt, N. Dey, and A. Joshi, Eds., Singapore: Springer, 2024, pp. 655–671.
- [199] A. Paya, S. Arroni, V. García-Díaz, and A. Gómez, "Apollon: A robust defense system against adversarial machine learning attacks in intrusion detection systems," *Comput. Secur.*, vol. 136, Jan. 2024, Art. no. 103546.
- [200] J. Chen, O. Gungor, Z. Shang, E. Li, and T. Rosing, "DYNAMITE: Dynamic defense selection for enhancing machine learning-based intrusion detection against adversarial attacks," in *Proc. IEEE Secur. Privacy Workshops (SPW)*, May 2025, pp. 213–219.
- [201] K. Barik and S. Misra, "IDS-anta: An open-source code with a defense mechanism to detect adversarial attacks for intrusion detection system," *Softw. Impacts*, vol. 21, Sep. 2024, Art. no. 100664.
- [202] X. Yuan, S. Han, W. Huang, H. Ye, X. Kong, and F. Zhang, "A simple framework to enhance the adversarial robustness of deep learning-based intrusion detection system," *Comput. Secur.*, vol. 137, Feb. 2024, Art. no. 103644.
- [203] Y. K. Beshah, S. L. Abebe, and H. M. Melaku, "Multi-stage adversarial defense for online DDoS attack detection system in IoT," *IEEE Access*, vol. 13, pp. 72657–72673, 2025.
- [204] S. R. Bommana, S. Veeramachaneni, S. E. Ahmed, and M. B. Srinivas, "Addressing adversarial attacks in IoT using deep learning AI models," *IEEE Access*, vol. 13, pp. 50437–50449, 2025.
- [205] S. Alahmed, Q. Alasad, M. M. Hammood, J.-S. Yuan, and M. Alawad, "Mitigation of black-box attacks on intrusion detection systems-based ML," *Computers*, vol. 11, no. 7, p. 115, Jul. 2022.
- [206] K. Roshan, A. Zafar, and S. B. Ul Haque, "A novel deep learning based model to defend network intrusion detection system against adversarial attacks," 2023, *arXiv:2308.00077*.
- [207] M. S. Haroon and H. M. Ali, "Adversarial training against adversarial attacks for machine learning-based intrusion detection systems," *Comput. Mater. Continua*, vol. 73, no. 2, pp. 3513–3527, 2022.
- [208] M. Pujari, Y. Pacheco, B. Cherukuri, and W. Sun, "A comparative study on the impact of adversarial machine learning attacks on contemporary intrusion detection datasets," *Social Netw. Comput. Sci.*, vol. 3, no. 5, p. 412, Aug. 2022.
- [209] A. F. Mukeri and D. P. Gaikwad, "Adversarial machine learning attacks and defenses in network intrusion detection systems," *Int. J. Wireless Microw. Technol.*, vol. 12, no. 1, pp. 12–21, Feb. 2022.
- [210] M. K. Roshan and A. Zafar, "Boosting robustness of network intrusion detection systems: A novel two phase defense strategy against untargeted white-box optimization adversarial attack," *Expert Syst. Appl.*, vol. 249, Sep. 2024, Art. no. 123567.
- [211] P. Sharma, A. Govind, S. Raj, and J. Mahur, "Adversarial machine learning for security: Experimental techniques for defending against AI-powered cyberattacks," *J. Sci. Innov. Adv. Res. (JSIAR)*, vol. 1, no. 1, 2025, Art. no. JSIAR-A-25-04004.
- [212] M. Schneider, D. Aspinall, and N. D. Bastian, "Evaluating model robustness to adversarial samples in network intrusion detection," in *Proc. IEEE Int. Conf. Big Data (Big Data)*, Dec. 2021, pp. 3343–3352.

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